

RoboSPA: Can VLA Models Go Beyond Simple Scenes and Short-Horizon Tasks?

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Abstract

Vision-Language-Action (VLA) models have shown promising progress in language-conditioned robotic manipulation. However, existing datasets and benchmarks mainly evaluate task completion under predefined settings, offering limited insight into model reasoning under increasing spatial and procedural complexity. We introduce **RoboSPA** (**Robot Spatial-Procedural Assessment**), a large-scale robotic manipulation dataset and benchmark for diagnosing embodied reasoning in VLA models. RoboSPA focuses on two core dimensions, Fine-Grained Spatial Reasoning and Long-Horizon Procedural Planning, covering 10 task categories and 56 base tasks. Each task is instantiated across five difficulty levels, yielding 280 variants with increasing spatial ambiguity and procedural complexity. We collect 527K trajectories across multiple embodiments and diverse scenes. Beyond binary success rate, RoboSPA introduces diagnostic metrics for more detailed evaluation. Experiments on representative VLA models show that current systems still struggle with complex spatial relations, precise low-level execution, and memory-intensive planning. These results establish RoboSPA as a challenging diagnostic benchmark for developing more capable, reliable, and generalizable embodied agents. Our data and code are available at <https://github.com/fanzhenxuan/RoboSPA>.

1 Introduction

Vision-Language-Action (VLA) models (Zitkovich et al., 2023; Kim et al., 2025; Black et al., 2025b; NVIDIA et al., 2025; Black et al., 2025a; Gao et al., 2026) have emerged as a promising paradigm for general-purpose embodied agents, integrating visual perception, language understanding, and action generation in a unified framework (Ma et al.,

2024; Kawaharazuka et al., 2025). With large-scale pretraining and multimodal alignment, these models show encouraging performance on manipulation tasks, especially in structured, short-horizon settings (Shao et al., 2025; Zhong et al., 2025).

However, deploying embodied agents in real-world scenarios requires capabilities beyond short-horizon instruction following in simple, well-structured scenes (Liu et al., 2025b; Zhang et al., 2025b; Wong et al., 2025; Yuan et al., 2025a; Dang et al., 2026). Everyday manipulation tasks pose three key challenges: (1) **Fine-grained target disambiguation**, requiring agents to identify the target among visually similar candidates through subtle spatial cues beyond category or color; (2) **Temporally extended task execution**, requiring multi-step execution under temporal constraints and accumulated errors; and (3) **Complexity-scalable embodied reasoning**, where increasing candidates, horizons, and environmental diversity amplify grounding and planning failures. These challenges call for benchmarks that evaluate embodied reasoning under increasing spatial and procedural complexity.

As shown in Table 1, existing robotic manipulation datasets and benchmarks leave several gaps for evaluating reasoning-oriented VLA models. First, fine-grained spatial reasoning is rarely evaluated explicitly, as prior works seldom test target disambiguation with subtle spatial cues. Second, long-horizon and step-level evaluation remain limited: LIBERO (Liu et al., 2023a) and RoboTwin 2.0 (Chen et al., 2026a) mainly focus on short-horizon tasks, while RoboCasa (Nasiriany et al., 2024) and MIKASA-Robo (Cherepanov et al., 2026) often lack step-level diagnosis. Third, most prior works lack controlled multi-level difficulty (Pumacay et al., 2024; Fei et al., 2026; Zhou et al., 2025), making it hard to analyze performance degradation as task complexity increases. Finally, dataset scale remains limited for broad

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Name	Categories	Tasks	Trajectories	Embodiments	Spatial Reasoning	Long-Horizon Tasks	Step-level Evaluation	Multi-Difficulty	Scene Diversity
BC-Z (Jang et al., 2022)	3	100+	26K	1	✗	✗	✗	✗	✗
RT-1 (Brohan et al., 2023)	8	700+	130K	1	✗	✓	✗	✗	✓
BridgeData V2 (Walke et al., 2023)	—	13	60.1K	1	✗	✗	✗	✗	✓
Open X-Embodiment (O’Neill et al., 2024)	527	160,266	1.4M	22	✗	✗	✗	✗	✓
DROID (Khazatsky et al., 2024)	—	86	76K	1	✗	✗	✗	✗	✓
RLBench (James et al., 2020)	—	100	—	1	✗	✓	✗	✗	✗
CALVIN (Mees et al., 2022)	—	34	N/A [†]	1	✗	✓	✓	✗	✓
LIBERO (Liu et al., 2023a)	4	130	6.5K	1	✗	✗	✗	✗	✓
ManiSkill2 (Gu et al., 2023)	4	20	N/A [‡]	1	✗	✗	✗	✗	✓
RoboCasa (Nasiriany et al., 2024)	—	100	~100K	1	✗	✓	✗	✗	✓
SimplerEnv (Li et al., 2025)	—	8	—	2	✗	✗	✗	✗	✓
MIKASA-Robo (Cherepanov et al., 2026)	12	32	32K	1	✗	✓	✗	✓	✗
RoboTwin 2.0 (Chen et al., 2026a)	—	50	≤137.5K	5	✗	✗	✗	✗	✓
LIBERO-Pro (Zhou et al., 2025)	4	40	—	1	✗	✗	✗	✗	✓
RMBench (Chen et al., 2026b)	2	9	450	1	✗	✓	✗	✗	✗
RoboSPA (Ours)	10	280	527K	5	✓	✓	✓	✓	✓

Table 1: **Comparison of representative robotic manipulation datasets and benchmarks with RoboSPA.** The upper and lower parts summarize datasets and benchmarks, respectively. [†]CALVIN reports approximately 24 hours of teleoperated demonstrations. [‡]ManiSkill2 reports over 4M demonstration frames.

reasoning evaluation. Most widely used datasets contain fewer than 200K trajectories, limiting task, embodiment, and scene coverage.

To address these limitations, we introduce **RoboSPA (Robot Spatial-Procedural Assessment)**, a large-scale robotic manipulation dataset and benchmark. To systematically study embodied reasoning under increasing task complexity, RoboSPA is guided by three core design principles:

- **Fine-Grained Spatial Reasoning.** As shown in Fig. 1, this component evaluates whether VLA models can ground instructions in complex spatial structures, such as geometric attributes, distances, cross-view cues, relations, and canonical indexing. This capability is essential for cluttered manipulation. To our knowledge, RoboSPA is the first VLA dataset to make fine-grained spatial reasoning a core evaluation dimension.
- **Long-Horizon Procedural Planning.** This component evaluates whether VLA models can execute manipulation tasks with multi-step decisions. As shown in Fig. 1, it covers repetitive procedures, order-constrained and order-free execution, composite coordination, and memory-intensive planning. These capabilities are crucial for real-world manipulation requiring temporal consistency and stepwise progress.
- **Multi-Level Hierarchical Evaluation.** This component measures how VLA model performance changes with increasing task complexity. Each base task is instantiated across five difficulty levels, enabling analysis of performance degradation under growing spatial complexity

and action horizons. Beyond task-level success, RoboSPA reports step-level progress for detailed diagnosis of model failures.

Together, these designs provide a controlled foundation for large-scale VLA data collection and evaluation. RoboSPA provides demonstrations in clean and domain-randomized scenes, covering 56 base tasks across 10 capability categories and five difficulty levels, yielding 280 variants. Each task supports step-level evaluation beyond final success rates. Across five embodiments, RoboSPA contains 527K trajectories and 997 hours of videos, forming a comprehensive benchmark for embodied reasoning under increasing task complexity.

We evaluate representative VLA models (Liu et al., 2025a; AgiBot-World-Contributors et al., 2025; Black et al., 2025a; Zheng et al., 2026) on RoboSPA and find that they struggle as task complexity increases. On the hardest tasks, all models achieve an average success rate below 25%, with some tasks dropping to 0%. Diagnostic analyses reveal failures in target grounding, low-level manipulation, long-horizon tracking, and memory-based reasoning, highlighting RoboSPA as a challenging diagnostic testbed for embodied reasoning.

2 Related Work

Vision-Language-Action Models. Large language models (LLMs) (Grattafiori et al., 2024; Yang et al., 2025; Cao et al., 2026; Fan et al., 2026) and multimodal large language models (MLLMs) (Liu et al., 2023b; Bai et al., 2025; Yuan et al., 2025b; Lin et al., 2025; Wang et al., 2026)

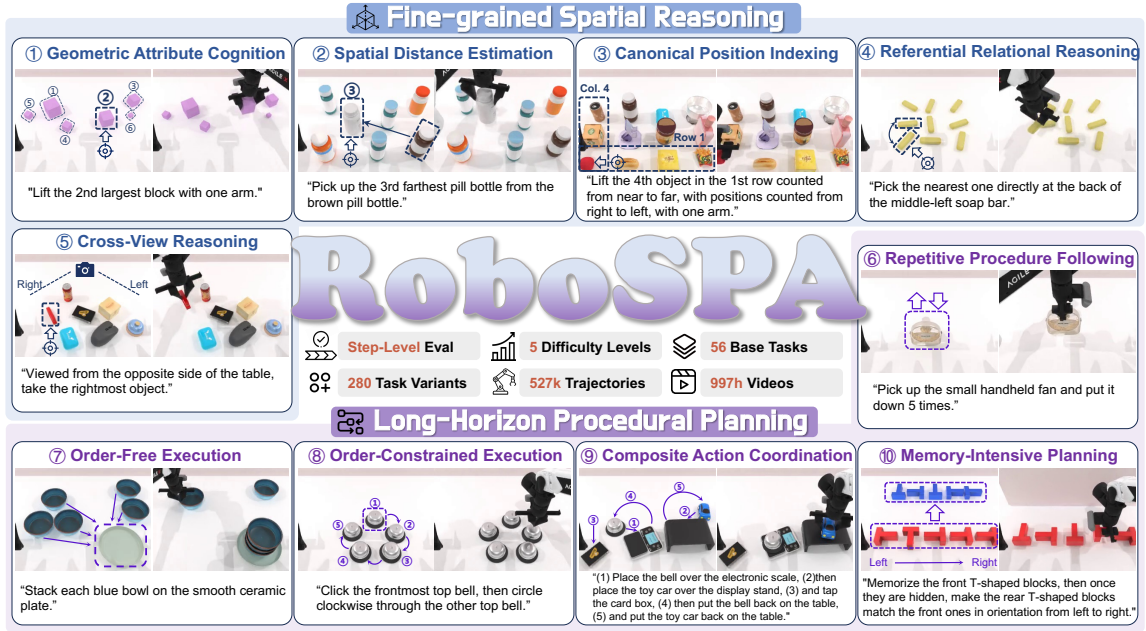
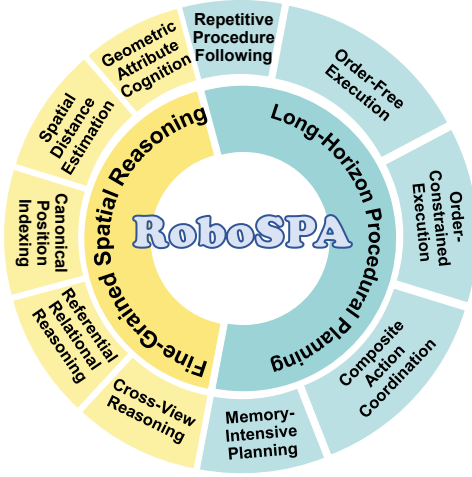


Figure 1: **Overview of RoboSPA.** RoboSPA centers on Fine-Grained Spatial Reasoning and Long-Horizon Procedural Planning, covering 10 task categories and 56 base tasks. Each task has 5 difficulty levels, yielding 280 variants. The dataset includes 527K trajectories across multiple embodiments and scene settings, supporting step-level evaluation of VLA limitations in spatial reasoning, long-horizon execution, and scalable embodied reasoning.

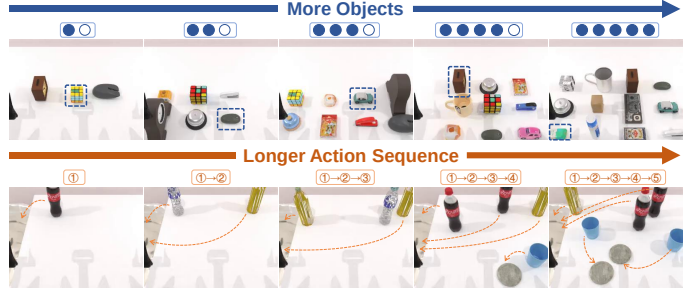
have demonstrated strong capabilities for understanding and reasoning over language and multimodal inputs, respectively. Building on these advances, VLA models map visual observations and language instructions to robot actions through a unified perception-to-control framework. Early methods, such as RT-1 (Brohan et al., 2023), RT-2 (Zitkovich et al., 2023), and OpenVLA (Kim et al., 2025), cast action prediction as autoregressive discrete token generation. Subsequent VLA models largely adopt diffusion or flow-matching policies for continuous action generation, as in RDT (Liu et al., 2025a) and π_0 (Black et al., 2025b). To improve cross-task generalization, VLA models such as $\pi_{0.5}$ (Black et al., 2025a) and GR00T N1 (NVIDIA et al., 2025) adopt hierarchical policies to bridge high-level language understanding with low-level motor execution, while other works enhance action and embodiment generalization through implicit action modeling and cross-embodiment data (AgiBot-World-Contributors et al., 2025; Zheng et al., 2026). From a different perspective, another line of work improves VLA models through spatial guidance, memory mechanisms, and future-frame prediction, further enhancing their capability and usability (Chen et al., 2025; Shi et al., 2026; Torne et al., 2026; Zhang et al., 2025d; Bi et al., 2026; Chen

et al., 2026c).

Robotic Manipulation Datasets and Benchmarks. The rapid development of VLA models has also accelerated the development of robotic manipulation datasets and evaluation benchmarks. Large-scale datasets, including BridgeData V2 (Walke et al., 2023), Open X-Embodiment (O’Neill et al., 2024), DROID (Khazatsky et al., 2024), and AgiBot World (AgiBot-World-Contributors et al., 2025), expand real-world, cross-embodiment manipulation coverage. Benchmark suites such as RL Bench (James et al., 2020), CALVIN (Mees et al., 2022), ManiSkill2 (Gu et al., 2023), and LIBERO (Liu et al., 2023a) provide structured settings for evaluating diverse skills, language-conditioned control, and sequential manipulation. Recent benchmarks further examine scene and capability generalization, sim-to-real transfer, bimanual manipulation, and memory-dependent tasks (Pumacay et al., 2024; Nasiriany et al., 2024; Li et al., 2025; Sedlacek et al., 2026; Garcia et al., 2025; Zhang et al., 2025c; Zhou et al., 2025; Zhang et al., 2025a; Chen et al., 2026a; Cherepanov et al., 2026; Chen et al., 2026b). However, they rarely systematically evaluate fine-grained spatial reasoning and long-horizon procedural planning with step-level diagnostics.



(a) Capability taxonomy



(b) Multi-difficulty task design



(c) Domain-randomized scenes

Figure 2: **Design of RoboSPA.** (a) Capability taxonomy with two dimensions and 10 categories. (b) Multi-difficulty task design through increasing object counts or action-sequence length. (c) Domain-randomized scenes with diverse layouts, distractors, lighting, textures, and tabletops.

RoboSPA addresses this gap with tasks specifically designed around these two capabilities.

3 RoboSPA

3.1 Overview

As shown in Fig. 1, we introduce RoboSPA, a large-scale robotic manipulation dataset and benchmark for evaluating reasoning-oriented VLA models. RoboSPA is built using the SAPIEN (Xiang et al., 2020) simulator and RoboTwin 2.0 (Chen et al., 2026a) framework. Each task is instantiated across multiple difficulty levels and scene settings.

3.2 Benchmark Construction

3.2.1 Capability Taxonomy

To systematically evaluate embodied reasoning in VLA models, we build a hierarchical capability taxonomy instead of treating manipulation tasks as isolated instances. As shown in Fig. 2(a), it includes two core dimensions: Fine-Grained Spatial Reasoning and Long-Horizon Procedural Planning.

Fine-Grained Spatial Reasoning. This dimension assesses instruction grounding in complex spatial configurations. Tasks require selecting the correct target through fine-grained spatial reasoning in complex scenes. It includes five categories:

- **Geometric Attribute Cognition (GAC):** Evaluates whether the model can identify and manipulate objects by geometric or shape-related attributes beyond category-level recognition.

- **Spatial Distance Estimation (SDE):** Measures whether the model can compare distances among objects or to a reference entity, reasoning about proximity, remoteness, and relative distance.
- **Canonical Position Indexing (CPI):** Evaluates whether the model can identify objects by row-column indices under different counting directions and spatial scanning orders.
- **Referential Relational Reasoning (RRR):** Assesses whether the model can locate targets through directional relations to reference objects, covering basic and compositional positions.
- **Cross-View Reasoning (CVR):** Examines whether the model can interpret spatial instructions from non-egocentric viewpoints by transforming spatial references across perspectives.

Long-Horizon Procedural Planning. This dimension evaluates a model’s ability to complete extended manipulation tasks with diverse actions, multiple subgoals, states, and procedural constraints. It also consists of five categories:

- **Repetitive Procedure Following (RPF):** Assesses whether the model can repeat a specified operation the required number of times while tracking progress and stopping correctly.
- **Order-Free Execution (OFE):** Measures whether the model can complete multiple

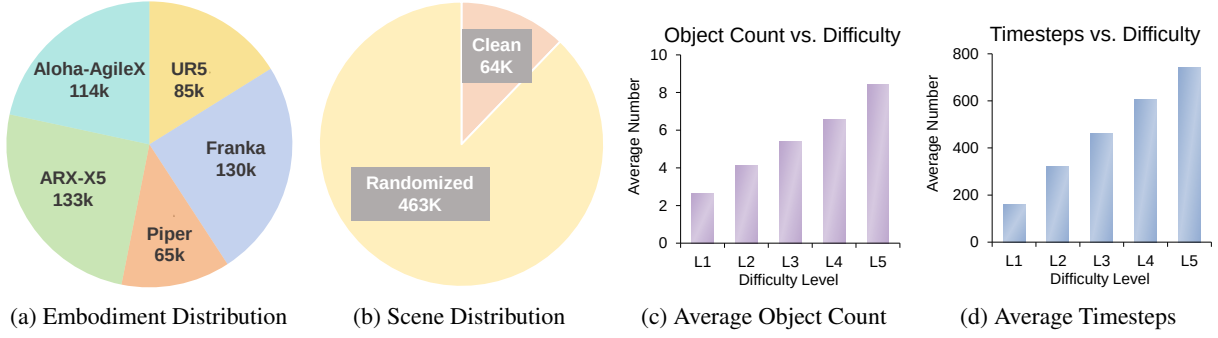


Figure 3: **Dataset statistics of RoboSPA.** (a) Trajectory distribution across five embodiments. (b) Trajectories across clean and domain-randomized scenes. (c) Average object count by difficulty for spatial reasoning tasks. (d) Average trajectory length by difficulty for long-horizon planning tasks.

subgoals in flexible order, covering all targets without omission or unnecessary repetition.

- **Order-Constrained Execution (OCE):** Evaluates whether the model can execute subgoals in a specified order, as correct actions in the wrong order can still fail.
- **Composite Action Coordination (CAC):** Examines whether the model can coordinate heterogeneous manipulation skills across multi-stage procedures with different action types.
- **Memory-Intensive Planning (MIP):** Tests whether the model can retain information and use it to guide later actions when cues must be recalled or are no longer available.

3.2.2 Task Suite Construction

Task Construction. Based on the proposed taxonomy, we design each task to instantiate a specific reasoning requirement. Tasks are defined by expert code specifying the scene, execution procedure, and success condition. We reuse part of RoboTwin 2.0 (Chen et al., 2026a) action primitives. Ten trained graduate students design the tasks, with each student responsible for one capability category. Each task is reviewed by two additional students for quality assurance.

Hierarchical Difficulty Design. To measure model performance under increasing reasoning burden, we instantiate each task across five difficulty levels. As shown in Fig. 2(b), difficulty is task-specific: spatial tasks increase candidate objects, while procedural tasks extend action sequences with more intermediate steps.

Instruction Design. For each task, we use GPT-5.2 (OpenAI, 2025) to generate 60 non-overlapping

instruction templates with diverse linguistic forms, using 50 for training and 10 for testing. These templates are instantiated with task-specific object descriptions and manually checked to ensure diversity under controlled semantics.

3.2.3 Data Collection

After task design, we execute expert code in simulation to collect trajectories in both clean and domain-randomized scenes. Clean scenes contain only task-relevant objects, while domain-randomized scenes vary clutter, textures, lighting, and tabletop configurations, as shown in Fig. 2(c), enabling evaluation of both task competence and scene robustness. We collect data across five embodiments: Aloha-AgileX, ARX-X5, Piper, Franka, and UR5.

3.2.4 Evaluation Metrics

To enable more diagnostic evaluation beyond *Success Rate (SR)*, we further introduce two dimension-specific complementary metrics.

For fine-grained spatial reasoning tasks, we introduce *Object-Normalized Target Accuracy (ONTA)*, inspired by Cohen’s kappa (Cohen, 1960), to fairly compare spatial reasoning ability under varying scene complexity. *ONTA* removes the object-count-induced chance baseline:

$$\text{ONTA}(n) = \left(\frac{\text{SR}(n) - 1/n}{1 - 1/n} \right) \times 100. \quad (1)$$

where n is the number of objects. *ONTA* is 100 for perfect target selection, 0 for random guessing, and negative for below-chance performance.

For long-horizon procedural planning tasks, we report *Progress Score (PS)* to capture partial completion. It is computed on a 0–100 scale:

$$\text{PS} = \left(\frac{1}{N} \sum_{i=1}^N \frac{c_i}{T} \right) \times 100. \quad (2)$$

Table 2: **Performance comparison of four baseline VLA models on RoboSPA.** We use *Success Rate (SR)* as the evaluation metric. Drop denotes the L1-to-L5 decrease. Blue shading marks the best model for each metric.

Task	RDT			GO-1			$\pi_{0.5}$			X-VLA		
	L1	L5	Drop	L1	L5	Drop	L1	L5	Drop	L1	L5	Drop
Fine-Grained Spatial Reasoning												
Geometric Attribute Cognition	10.0	4.8	5.2	27.8	11.5	16.3	42.8	12.5	30.3	57.8	25.3	32.5
Spatial Distance Estimation	15.2	9.4	5.8	20.2	11.6	8.6	35.0	23.4	11.6	41.4	34.6	6.8
Canonical Position Indexing	15.0	8.6	6.4	22.0	7.2	14.8	44.0	23.6	20.4	31.8	8.8	23.0
Referential Relational Reasoning	4.4	4.4	0.0	17.2	6.6	10.6	33.2	9.8	23.4	40.4	14.2	28.2
Cross-View Reasoning	3.0	3.4	-0.4	17.4	16.2	1.2	41.2	37.6	3.6	38.2	36.6	1.6
<i>Average</i>	9.5	6.1	3.4	20.9	10.6	10.3	39.2	21.4	17.8	41.9	23.9	18.0
Long-Horizon Procedural Planning												
Repetitive Procedure Following	40.8	35.0	5.8	31.8	23.8	8.0	80.3	65.0	15.3	34.8	22.5	12.3
Order-Free Execution	20.3	0.8	19.5	41.1	5.6	35.5	85.8	26.1	59.7	86.0	43.6	42.4
Order-Constrained Execution	13.6	0.1	13.5	41.9	3.1	38.8	77.4	12.0	65.4	70.0	7.7	62.3
Composite Action Coordination	27.5	2.4	25.1	24.2	2.6	21.6	58.6	12.5	46.1	58.9	5.9	53.0
Memory-Intensive Planning	18.2	0.0	18.2	7.6	0.0	7.6	54.0	0.0	54.0	44.6	0.0	44.6
<i>Average</i>	24.1	7.7	16.4	29.3	7.0	22.3	71.2	23.1	48.1	58.8	15.9	42.9
Overall Benchmark												
<i>Overall Average</i>	16.8	6.9	9.9	25.1	8.8	16.3	55.2	22.3	32.9	50.4	19.9	30.5

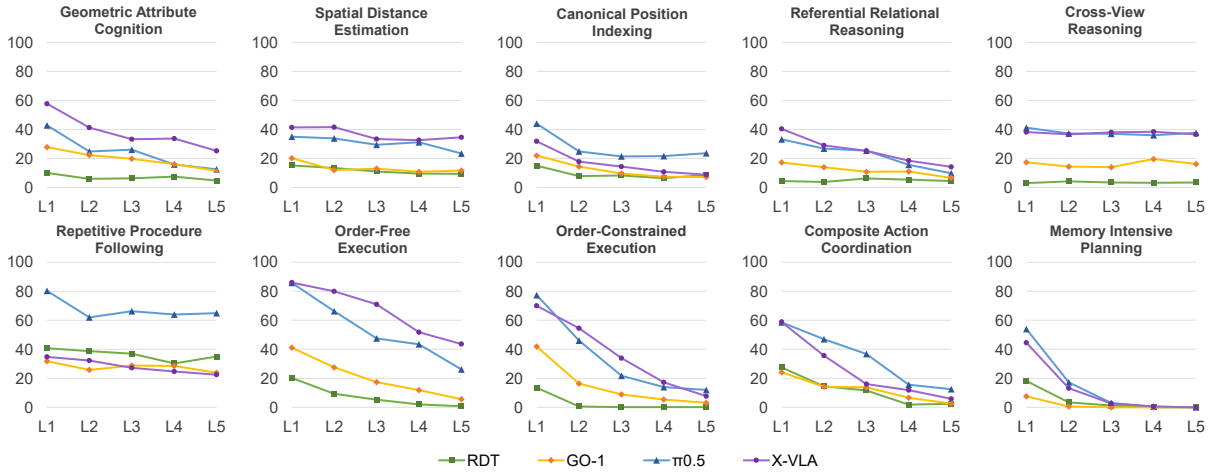


Figure 4: Success rate trends of four baseline VLA models across 10 task categories over five difficulty levels.

where T is the total number of subtasks, and c_i is the number completed in episode i .

3.3 Benchmark Statistics

RoboSPA contains **56** base tasks across two capability dimensions. Each base task is instantiated at **5** difficulty levels, yielding **280** task variants. In total, we collect approximately **527K** trajectories, corresponding to over **997** hours of interaction videos and **108M** timesteps. More dataset statistics are provided in the Appendix C. Fig. 3(a) presents the trajectory distribution across the **5** robotic embodiments, while Fig. 3(b) shows the trajectory distribution across clean and domain-randomized scenes. Fig. 3(c) and Fig. 3(d) further illustrate the hierar-

chical difficulty design along the two dimensions, respectively.

4 Experiments

4.1 Experimental Setup

We evaluate four representative VLA models, RDT (Liu et al., 2025a), GO-1 (AgiBot-World-Contributors et al., 2025), $\pi_{0.5}$ (Black et al., 2025a), and X-VLA (Zheng et al., 2026), using the clean-scene data collected with the Aloha-AgileX embodiment for both training and evaluation. All models are trained and evaluated under a single-task setting. For each base task, we train a separate model using data from all five difficulty levels and evaluate

Table 3: Comparison of *Object-Normalized Target Accuracy (ONTA)* on fine-grained spatial reasoning tasks.

Task	RDT			GO-1			$\pi_{0.5}$			X-VLA		
	L1	L3	L5	L1	L3	L5	L1	L3	L5	L1	L3	L5
Geometric Attribute Cognition	-80.0	-25.0	-14.3	-44.5	-7.0	-6.2	-14.5	1.3	-5.0	15.5	11.0	10.3
Spatial Distance Estimation	-27.2	-9.1	-4.6	-19.7	-6.7	-2.1	2.5	13.6	11.6	12.1	18.2	24.4
Canonical Position Indexing	-27.5	-4.9	0.3	-17.0	-3.3	-1.2	16.0	10.2	16.7	-2.3	2.2	0.5
Referential Relational Reasoning	-91.2	-25.1	-7.6	-65.6	-18.9	-5.1	-33.6	0.5	-1.5	-19.2	0.3	3.5
Cross-View Reasoning	-45.5	-28.8	-28.8	-23.9	-14.7	-11.7	11.8	16.0	16.8	7.3	17.3	15.5
<i>Average</i>	-54.3	-18.6	-11.0	-34.1	-10.1	-5.3	-3.6	8.3	7.7	2.7	9.8	10.8

Table 4: Performance comparison in terms of *Progress Score (PS)* on long-horizon procedural planning tasks.

Task	RDT			GO-1			$\pi_{0.5}$			X-VLA		
	L1	L3	L5	L1	L3	L5	L1	L3	L5	L1	L3	L5
Repetitive Procedure Following	40.8	46.3	44.5	31.8	35.0	32.0	80.3	78.7	77.3	34.8	34.8	31.9
Order-Free Execution	20.3	33.0	24.9	41.1	39.7	25.7	85.8	75.0	62.1	86.0	84.4	73.9
Order-Constrained Execution	13.6	25.3	14.7	41.9	24.1	14.8	77.4	44.3	28.0	70.0	51.1	25.1
Composite Action Coordination	27.5	32.5	26.5	24.2	26.2	22.7	58.6	60.5	48.0	58.9	49.0	40.1
Memory-Intensive Planning	18.2	5.0	3.6	7.6	4.9	6.0	54.0	18.7	11.4	44.6	12.2	7.3
<i>Average</i>	24.1	28.4	22.8	29.3	26.0	20.2	71.2	55.4	45.4	58.8	46.3	35.7

it independently at each level. During evaluation, each model is tested with 100 rollout trials per task variant.

4.2 Main Results

Table 2 reports the *Success Rates (SR)* of four representative VLA models. Overall, model performance declines from L1 to L5, indicating limited ability to reason under complex task settings.

Model Comparison. Among the evaluated models, $\pi_{0.5}$ performs best overall, but reaches only 22.3% success at L5, followed by X-VLA with 19.9%. Both drop by over 30 percentage points as difficulty increases. GO-1 and RDT perform worse, achieving only 8.8% and 6.9% success at L5, respectively. These results show that stronger models handle easier cases better, but all models still lack robust reasoning under complex conditions.

Fine-Grained Spatial Reasoning. X-VLA shows the best performance in this dimension, but its success rate still drops from 41.9% at L1 to only 23.9% at L5. This decline shows that increasing spatial complexity substantially weakens target grounding. Models struggle particularly with Canonical Position Indexing (CPI) and Referential Relational Reasoning (RRR), indicating that current VLA models remain unreliable when target selection depends on fine-grained spatial relations rather than simple object recognition.

Long-Horizon Procedural Planning. For Long-Horizon Procedural Planning, the performance drop is even more pronounced, with the best-performing model, $\pi_{0.5}$, decreasing from 71.2% at L1 to 23.1% at L5. Models perform best on Repetitive Procedure Following (RPF), while the other four categories show substantially lower performance. Notably, all models fail on the hardest Memory-Intensive Planning (MIP), revealing severe limitations in maintaining task-relevant memory over extended horizons.

Fig. 4 further shows that performance drops from L1 to L5, exposing failures in spatial discrimination and long-horizon tracking.

4.3 Diagnostic Evaluation

Object-Normalized Target Accuracy. Table 3 compares models using the *ONTA* metric. Overall, all models achieve low scores. RDT and GO-1 obtain negative *ONTA* on most categories and difficulty levels, showing little spatial reasoning beyond random selection. $\pi_{0.5}$ and X-VLA show fewer negative scores and achieve better average *ONTA*, but their overall scores remain close to the random-selection baseline. These results suggest that current VLA models still struggle to identify the correct target when success depends on fine-grained spatial cues.

Progress Score. Table 4 reports *Progress Score* for long-horizon procedural planning tasks, provid-

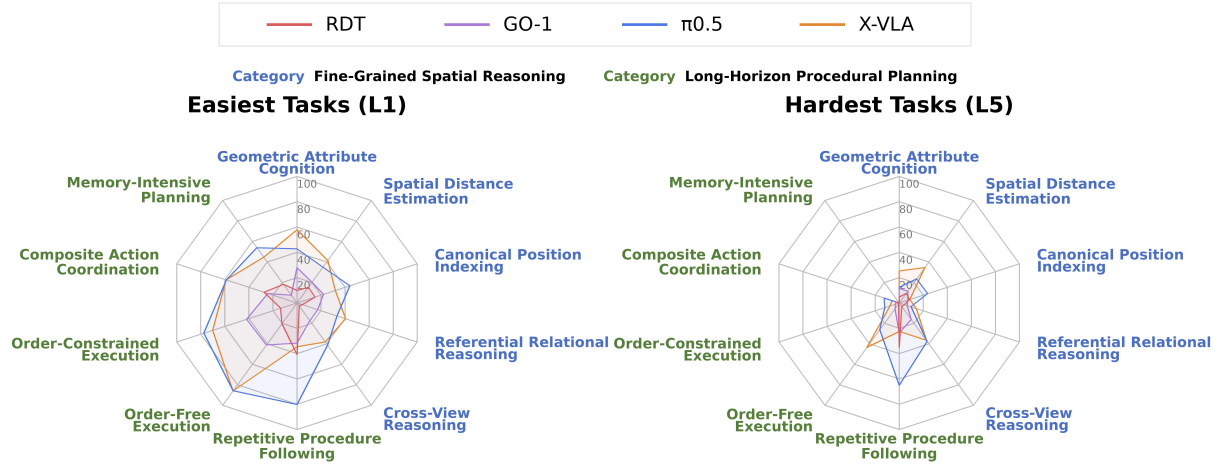


Figure 5: **Category-wise performance of four baseline VLA models on RoboSPA.** Left: *Success Rate* across 10 capability categories at L1. Right: *Success Rate* across 10 capability categories at L5.

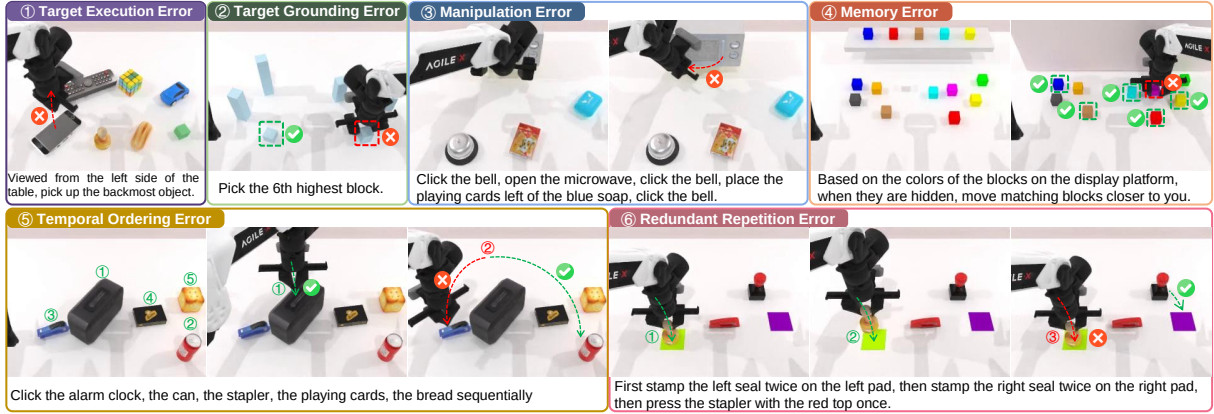


Figure 6: **Representative failure modes observed on RoboSPA.** Failure modes vary across different RoboSPA task categories, showing that our benchmark can distinguish model weaknesses along different capability dimensions.

ing a progress-based view beyond binary *Success Rate*. The results show that *Progress Score* is generally higher than final *Success Rate*, indicating that models can often execute part of a procedure but fail due to accumulated errors, incorrect ordering, or incomplete later-stage execution. $\pi_{0.5}$ achieves the strongest average completion rate, followed by X-VLA. Nevertheless, Memory-Intensive Planning (MIP) remains weak at higher difficulty levels, indicating that current models still have limited memory ability during long-horizon execution.

4.4 Qualitative Results

Fig. 5 visualizes category-wise model performance across task difficulty. The radar plots show that model performance is highly uneven across categories and shrinks markedly under harder settings. This reveals persistent limitations in handling increasingly complex embodied reasoning tasks.

4.5 Failure Analysis

We further analyze model failures. Fig. 6 presents six representative failure modes, with quantitative details provided in the Appendix A.

Observed Failure Modes. For fine-grained spatial reasoning, target execution error denotes failed manipulation after correct grounding, while target grounding error denotes selecting a distractor. For long-horizon procedural planning, manipulation error refers to failed intermediate execution, memory error to forgetting earlier task-relevant information, temporal ordering error to executing subtasks in the wrong order, and redundant repetition error to repeating completed actions instead of progressing.

Implications for Future Models. These findings suggest several directions for future VLA models. Improving spatial performance requires stronger object-centric grounding and relation awareness, together with reliable low-level execution. Long-

horizon tasks require stronger progress tracking, memory, and execution monitoring to maintain task state. These abilities help models follow temporal constraints and reduce ordering errors, redundant repetitions, and cascading failures.

5 Conclusion

We present RoboSPA, a reasoning-focused, large-scale robotic manipulation dataset and benchmark. Centered on Fine-Grained Spatial Reasoning and Long-Horizon Procedural Planning, RoboSPA covers ten task categories, five difficulty levels, multiple embodiments, and diverse scenes, with fine-grained metrics for spatial grounding and subtask completion. Experiments show that existing VLA models still struggle with complex spatial relations, precise execution, and memory-intensive planning, especially at higher difficulty levels. We hope RoboSPA can advance the development of more capable and reliable embodied agents.

Limitations

Although our dataset covers diverse fine-grained spatial reasoning and long-horizon procedural planning tasks, several limitations remain. First, all tasks are constructed in simulation, and the sim-to-real gap may limit direct transfer to physical robots. Second, our benchmark focuses on tabletop manipulation, which may not fully capture the diversity and open-endedness of real-world environments. Third, while we include multiple robotic embodiments and domain-randomized scenes, broader settings such as deformable object manipulation, human-robot interaction, and open-ended task instructions remain underexplored.

Ethics Statement

Scope and Safety. RoboSPA is designed to diagnose and evaluate VLA models in simulated robotic manipulation environments, rather than to serve as a directly deployable real-world robotic system. Data collection is conducted entirely in simulation and does not involve real-world robot deployment. Since models evaluated or developed with RoboSPA may eventually be transferred to physical environments, such deployment should include appropriate safety constraints, human oversight, and task-specific validation.

Data Content and Privacy. RoboSPA does not involve human-subject experiments, personal data, or

real-world private data, and thus poses no privacy risks to human participants. We checked the task instructions, object categories, scene assets, and collected demonstrations to ensure that the dataset does not contain personally identifying information or offensive, hateful, or explicit content. Therefore, no additional anonymization is required.

Artifact Licensing. RoboSPA builds upon third-party open-source resources. We use the SAPIEN simulator, which is released under the Apache License 2.0, and reuse code and 3D assets from RoboTwin 2.0 and RMBench, which are released under the MIT License. Baseline models and pre-trained checkpoints used in our experiments are obtained from their official releases and used in accordance with their original license terms. These third-party artifacts are used consistently with their intended research purposes for simulation-based robotic manipulation, benchmark construction, data generation, and evaluation. We retain the original copyright and license notices of all third-party resources. The artifacts introduced by RoboSPA, including task definitions, benchmark code, evaluation scripts, and collected demonstration data, will be released under the MIT License.

Responsible Use. We acknowledge that simulator design, task selection, object categories, and embodiment choices may introduce biases and limit transferability, which we mitigate through multiple embodiments, diverse task categories, controlled difficulty levels, and both clean and domain-randomized scenes. We also encourage responsible use of RoboSPA, including efficient training, transparent reporting of computational costs, and consideration of environmental impact.

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References

AgiBot-World-Contributors, Qingwen Bu, Jisong Cai, Li Chen, Xiuqi Cui, Yan Ding, Siyuan Feng, Shenyuan Gao, Xindong He, Xuan Hu, Xu Huang, Shu Jiang, Yuxin Jiang, Cheng Jing, Hongyang Li, Jialu Li, Chiming Liu, Yi Liu, Yuxiang Lu, and 33 others. 2025. [Agibot world colosseum: A large-scale](#)

- manipulation platform for scalable and intelligent embodied systems. *Preprint*, arXiv:2503.06669.
- Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, Wenbin Ge, Zhi-fang Guo, Qidong Huang, Jie Huang, Fei Huang, Binyuan Hui, Shutong Jiang, Zhaohai Li, Mingsheng Li, and 45 others. 2025. [Qwen3-vl technical report](#). *Preprint*, arXiv:2511.21631.
- Hongzhe Bi, Hengkai Tan, Shenghao Xie, Zeyuan Wang, Shuhe Huang, Haitian Liu, Ruowen Zhao, Yao Feng, Chendong Xiang, Yinze Rong, Hongyan Zhao, Hanyu Liu, Zhizhong Su, Lei Ma, Hang Su, and Jun Zhu. 2026. Motus: A unified latent action world model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 35101–35113.
- Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, brian ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, and 16 others. 2025a. [\$\pi_{0.5}\$: a vision-language-action model with open-world generalization](#). In *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 17–40. PMLR.
- Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, and 6 others. 2025b. [\$\pi_0\$: A Vision-Language-Action Flow Model for General Robot Control](#). In *Proceedings of Robotics: Science and Systems*, Los Angeles, CA, USA.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, and 32 others. 2023. [RT-1: Robotics Transformer for Real-World Control at Scale](#). In *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea.
- Jie Cao, Tianwei Lin, Zhenxuan Fan, Bo Yuan, Ziyuan Zhao, Rolan Yan, Wenqiao Zhang, and Siliang Tang. 2026. [Draft-thinking: Learning efficient reasoning in long chain-of-thought llms](#). *Preprint*, arXiv:2603.00578.
- Tianxing Chen, Zhanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, Yiheng Ge, Zhenyu Gu, Weiliang Deng, Yubin Guo, Tian Nian, Xuanbing Xie, Qiangyu Chen, Kailun Su, Tianling Xu, Guodong Liu, Mengkang Hu, and 7 others. 2026a. [Robotwin 2.0: A scalable data generator and benchmark with strong domain randomization for robust bimanual robotic manipulation](#). In *Forty-third International Conference on Machine Learning*.
- Tianxing Chen, Yuran Wang, Mingleyang Li, Yan Qin, Hao Shi, Zixuan Li, Yifan Hu, Yingsheng Zhang, Kaixuan Wang, Yue Chen, Hongcheng Wang, Junjie Wang, Tianhang Yang, Renjing Xu, Ruihai Wu, Yao Mu, Yaodong Yang, Hao Dong, and Ping Luo. 2026b. [Rmbench: Memory-dependent robotic manipulation benchmark with insights into policy design](#). *Preprint*, arXiv:2603.01229.
- Xinyi Chen, Yilun Chen, Yanwei Fu, Ning Gao, Ji-aya Jia, Weiyang Jin, Hao Li, Yao Mu, Jiang-miao Pang, Yu Qiao, Yang Tian, Bin Wang, Bolun Wang, Fangjing Wang, Hanqing Wang, Tai Wang, Ziqin Wang, Xueyuan Wei, Chao Wu, and 10 others. 2025. [Internvl-m1: A spatially guided vision-language-action framework for generalist robot policy](#). *Preprint*, arXiv:2510.13778.
- Yi Chen, Yuying Ge, Hui Zhou, Mingyu Ding, Yixiao Ge, and Xihui Liu. 2026c. [Dial: Decoupling intent and action via latent world modeling for end-to-end vla](#). *Preprint*, arXiv:2603.29844.
- Egor Cherepanov, Nikita Kachaev, Alexey Kovalev, and Aleksandr Panov. 2026. [Memory, benchmark & robots: A benchmark for solving complex tasks with reinforcement learning](#). In *The Fourteenth International Conference on Learning Representations*.
- Jacob Cohen. 1960. [A coefficient of agreement for nominal scales](#). *Educational and Psychological Measurement*, 20(1):37–46.
- Ronghao Dang, Jiayan Guo, Bohan Hou, Sicong Leng, Kehan Li, Xin Li, Jiangpin Liu, Yunxuan Mao, Zhikai Wang, Yuqian Yuan, Minghao Zhu, Xiao Lin, Yang Bai, Qian Jiang, Yaxi Zhao, Minghua Zeng, Junlong Gao, Yuming Jiang, Jun Cen, and 7 others. 2026. [Rynnbrian: Open embodied foundation models](#). *Preprint*, arXiv:2602.14979.
- Zhenxuan Fan, Jie Cao, Yang Dai, Zheqi Lv, Wenqiao Zhang, Zhongle Xie, Peng LU, and Beng Chin Ooi. 2026. [Ctrlcot: Dual-granularity chain-of-thought compression for controllable reasoning](#). *Preprint*, arXiv:2601.20467.
- Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang, Zhaoye Fei, Jinlan Fu, Jingjing Gong, and Xipeng Qiu. 2026. [Libero-plus: A progressive robustness benchmark for visual-language-action models](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 38574–38583.
- Mingjian Gao, Wenqiao Zhang, Yuqian Yuan, Yang Dai, Binhe Yu, Zheqi Lv, Haoyu Zheng, Jiaqi Zhu, Zhiqi Ge, Zixuan Wan, Siliang Tang, and Yueting Zhuang. 2026. [Visualthink-vla: Visual intermediate reasoning for effective and low-latency vision-language-action policies](#). *Preprint*, arXiv:2605.30011.

- Ricardo Garcia, Shizhe Chen, and Cordelia Schmid. 2025. Towards generalizable vision-language robotic manipulation: A benchmark and llm-guided 3d policy. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 8996–9002. IEEE.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. *The llama 3 herd of models*. Preprint, arXiv:2407.21783.
- Jiayuan Gu, Fanbo Xiang, Xuanlin Li, Zhan Ling, Xiqiang Liu, Tongzhou Mu, Yihe Tang, Stone Tao, Xinyue Wei, Yunchao Yao, Xiaodi Yuan, Pengwei Xie, Zhiao Huang, Rui Chen, and Hao Su. 2023. *Maniskill2: A unified benchmark for generalizable manipulation skills*. In *The Eleventh International Conference on Learning Representations*.
- Stephen James, Zicong Ma, David Rovick Arrojo, and Andrew J. Davison. 2020. *Rlbench: The robot learning benchmark & learning environment*. *IEEE Robotics and Automation Letters*, 5(2):3019–3026.
- Eric Jang, Alex Irpan, Mohi Khansari, Daniel Kappler, Frederik Ebert, Corey Lynch, Sergey Levine, and Chelsea Finn. 2022. Bc-z: Zero-shot task generalization with robotic imitation learning. In *conference on Robot Learning*, pages 991–1002. PMLR.
- Kento Kawaharazuka, Jihoon Oh, Jun Yamada, Ingmar Posner, and Yuke Zhu. 2025. *Vision-language-action models for robotics: A review towards real-world applications*. *IEEE Access*, 13:162467–162504.
- Alexander Khazatsky, Karl Pertsch, Suraj Nair, Ashwin Balakrishna, Sudeep Dasari, Siddharth Karamcheti, Soroush Nasiriany, Mohan Kumar Srirama, Lawrence Yunliang Chen, Kirsty Ellis, Peter David Fagan, Joey Hejna, Masha Itkina, Marion Lepert, Yecheng Jason Ma, Patrick Tree Miller, Jimmy Wu, Suneel Belkale, Shivin Dass, and 79 others. 2024. *DROID: A large-scale in-the-wild robot manipulation dataset*. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan P Foster, Pannag R Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. 2025. *Openvla: An open-source vision-language-action model*. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 2679–2713. PMLR.
- Xuanlin Li, Kyle Hsu, Jiayuan Gu, Oier Mees, Karl Pertsch, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. 2025. *Evaluating real-world robot manipulation policies in simulation*. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 3705–3728. PMLR.
- Tianwei Lin, Wenqiao Zhang, Sijing Li, Yuqian Yuan, Binhe Yu, Haoyuan Li, Wanggui He, Hao Jiang, Mengze Li, Xiaohui Song, Siliang Tang, Jun Xiao, Hui Lin, Yueting Zhuang, and Beng Chin Ooi. 2025. *Healthgpt: A medical large vision-language model for unifying comprehension and generation via heterogeneous knowledge adaptation*. In *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net.
- Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. 2023a. *Libero: Benchmarking knowledge transfer for lifelong robot learning*. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*.
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023b. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916.
- Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. 2025a. *RDT-1b: a diffusion foundation model for bimanual manipulation*. In *The Thirteenth International Conference on Learning Representations*.
- Yang Liu, Weixing Chen, Yongjie Bai, Xiaodan Liang, Guanbin Li, Wen Gao, and Liang Lin. 2025b. *Aligning cyber space with physical world: A comprehensive survey on embodied ai*. *IEEE/ASME Transactions on Mechatronics*, 30(6):7253–7274.
- Yueen Ma, Zixing Song, Yuzheng Zhuang, Jianye Hao, and Irwin King. 2024. A survey on vision-language-action models for embodied ai. *arXiv preprint arXiv:2405.14093*.
- Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. 2022. *Calvin: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks*. *IEEE Robotics and Automation Letters*, 7(3):7327–7334.
- Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. 2024. *RoboCasa: Large-Scale Simulation of Household Tasks for Generalist Robots*. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands.
- NVIDIA, :, Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi "Jim" Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, Joel Jang, Zhenyu Jiang, Jan Kautz, Kaushil Kundalia, Lawrence Lao, Zhiqi Li, Zongyu Lin, and

- 24 others. 2025. [Gr00t n1: An open foundation model for generalist humanoid robots](#). *Preprint*, arXiv:2503.14734.
- OpenAI. 2025. Introducing gpt-5.2. <https://openai.com/index/introducing-gpt-5-2/>.
- Abby O'Neill, Abdul Rehman, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, Ajinkya Jain, Albert Tung, Alex Bewley, Alex Herzog, Alex Irpan, Alexander Khazatsky, Anant Rai, Anchit Gupta, Andrew Wang, Anikait Singh, and 260 others. 2024. [Open x-embodiment: Robotic learning datasets and rt-x models : Open x-embodiment collaboration0](#). In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6892–6903.
- Wilbert Pumacay, Ishika Singh, Jiafei Duan, Ranjay Krishna, Jesse Thomason, and Dieter Fox. 2024. [The Colosseum: A Benchmark for Evaluating Generalization for Robotic Manipulation](#). In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands.
- Martin Sedlacek, Pavlo Yefanov, Georgy Ponimatkin, Jai Bardhan, Simon Pilc, Mederic Fourmy, Evangelos Kazakos, Cees G. M. Snoek, Josef Sivic, and Vladimir Petrik. 2026. [Realm: A real-to-sim validated benchmark for generalization in robotic manipulation](#). *IEEE Robotics and Automation Letters*, 11(7):8315–8322.
- Rui Shao, Wei Li, Lingsen Zhang, Renshan Zhang, Zhiyang Liu, Ran Chen, and Liqiang Nie. 2025. Large vlm-based vision-language-action models for robotic manipulation: A survey. *arXiv preprint arXiv:2508.13073*.
- Hao Shi, Bin Xie, Yingfei Liu, Lin Sun, Fengrong Liu, Tiancai Wang, Erjin Zhou, Haoqiang Fan, Xiangyu Zhang, and Gao Huang. 2026. [MemoryVLA: Perceptual-cognitive memory in vision-language-action models for robotic manipulation](#). In *The Fourteenth International Conference on Learning Representations*.
- Marcel Torne, Karl Pertsch, Homer Walke, Kyle Vedder, Suraj Nair, Brian Ichter, Allen Z. Ren, Haohuan Wang, Jiaming Tang, Kyle Stachowicz, Karan Dhabalia, Michael Equi, Quan Vuong, Jost Tobias Springenberg, Sergey Levine, Chelsea Finn, and Danny Driess. 2026. [Mem: Multi-scale embodied memory for vision language action models](#). *Preprint*, arXiv:2603.03596.
- Homer Rich Walke, Kevin Black, Tony Z. Zhao, Quan Vuong, Chongyi Zheng, Philippe Hansen-Estruch, Andre Wang He, Vivek Myers, Moo Jin Kim, Max Du, Abraham Lee, Kuan Fang, Chelsea Finn, and Sergey Levine. 2023. [Bridgedata v2: A dataset for robot learning at scale](#). In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 1723–1736. PMLR.
- Zhuonan Wang, Zhenxuan Fan, Siwen Tan, Yu Zhong, Yuqian Yuan, Haoyuan Li, Hao Jiang, Wenqiao Zhang, Feifei Shao, Hongwei Wang, and Jun Xiao. 2026. [MAU-GPT: enhancing multi-type industrial anomaly understanding via anomaly-aware and generalist experts adaptation](#). In *Fortieth AAAI Conference on Artificial Intelligence, Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence, Sixteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2026, Singapore, January 20-27, 2026*, pages 26787–26795. AAAI Press.
- Lik Hang Kenny Wong, Xueyang Kang, Kaixin Bai, and Jianwei Zhang. 2025. A survey of robotic navigation and manipulation with physics simulators in the era of embodied ai. *arXiv preprint arXiv:2505.01458*.
- Fanbo Xiang, Yuzhe Qin, Kaichun Mo, Yikuan Xia, Hao Zhu, Fangchen Liu, Minghua Liu, Hanxiao Jiang, Yifu Yuan, He Wang, Li Yi, Angel X. Chang, Leonidas J. Guibas, and Hao Su. 2020. Sapien: A simulated part-based interactive environment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 41 others. 2025. [Qwen3 technical report](#). *Preprint*, arXiv:2505.09388.
- Yuqian Yuan, Ronghao Dang, long li, Wentong Li, Dian Jiao, Xin Li, Deli Zhao, Fan Wang, Wenqiao Zhang, Jun Xiao, and Yueting Zhuang. 2025a. [Eoc-bench: Can mllms identify, recall, and forecast objects in an egocentric world?](#) In *Advances in Neural Information Processing Systems*, volume 38, Main Conference. Curran Associates, Inc.
- Yuqian Yuan, Hang Zhang, Wentong Li, Zesen Cheng, Boqiang Zhang, Long Li, Xin Li, Deli Zhao, Wenqiao Zhang, Yueting Zhuang, Jianke Zhu, and Lidong Bing. 2025b. [Vidorefer suite: Advancing spatial-temporal object understanding with video llm](#). In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18970–18980.
- Borong Zhang, Jiahao Li, Jiachen Shen, Yishuai Cai, Yuhao Zhang, Yuanpei Chen, Juntao Dai, Jiaming Ji, and Yaodong Yang. 2025a. [VLA-arena: An open-source framework for benchmarking vision-language-action models](#). *arXiv preprint arXiv:2512.22539*.
- Kun Zhang, Peng Yun, Jun Cen, Junhao Cai, Didi Zhu, Hangjie Yuan, Chao Zhao, Tao Feng, Michael Yu Wang, Qifeng Chen, Jia Pan, Wei Zhang, Bo Yang, and Hua Chen. 2025b. [Generative artificial intelligence in robotic manipulation: A survey](#). *Preprint*, arXiv:2503.03464.
- Shiduo Zhang, Zhe Xu, Peiju Liu, Xiaopeng Yu, Yuan Li, Qinghui Gao, Zhaoye Fei, Zhangyue Yin, Zuxuan Wu, Yu-Gang Jiang, and Xipeng Qiu. 2025c.

Vlabench: A large-scale benchmark for language-conditioned robotics manipulation with long-horizon reasoning tasks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11142–11152.

Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, XinQiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. 2025d. [Dreamvla: A vision-language-action model dreamed with comprehensive world knowledge](#). In *Advances in Neural Information Processing Systems*.

Jinliang Zheng, Jianxiong Li, Zhihao Wang, Dongxiu Liu, Xirui Kang, Yuchun Feng, Yinan Zheng, Jiayin Zou, Yilun Chen, Jia Zeng, Tai Wang, Ya-Qin Zhang, Jingjing Liu, and Xianyu Zhan. 2026. [X-VLA: Soft-prompted transformer as scalable cross-embodiment vision-language-action model](#). In *The Fourteenth International Conference on Learning Representations*.

Yifan Zhong, Fengshuo Bai, Shaofei Cai, Xuchuan Huang, Zhang Chen, Xiaowei Zhang, Yuanfei Wang, Shaoyang Guo, Tianrui Guan, Ka Nam Lui, Zhiqian Qi, Yitao Liang, Yuanpei Chen, and Yaodong Yang. 2025. [A survey on vision-language-action models: An action tokenization perspective](#). *Preprint*, arXiv:2507.01925.

Xueyang Zhou, Yangming Xu, Guiyao Tie, Yongchao Chen, Guowen Zhang, Duanfeng Chu, Pan Zhou, and Lichao Sun. 2025. [Libero-pro: Towards robust and fair evaluation of vision-language-action models beyond memorization](#). *Preprint*, arXiv:2510.03827.

Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, and 35 others. 2023. [Rt-2: Vision-language-action models transfer web knowledge to robotic control](#). In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR.

Appendix

In this document, we provide additional details about RoboSPA. The appendix is organized as follows:

- §A: Detailed Failure Analysis.
- §B: Training and Evaluation Details.
- §C: Dataset Details.
- §D: Basic Grounding Tasks.
- §E: Task List and Descriptions.

- §F: Detailed Experimental Results.
- §G: Task Visualizations.

A Detailed Failure Analysis

A.1 Analysis Setup

We perform a detailed error analysis on the two best-performing models, $\pi_{0.5}$ and X-VLA. For each model, we analyze approximately 1,120 evaluation videos, sampling up to 20 error examples per L5 task; if fewer than 20 errors occur, we include all available examples. For fine-grained spatial reasoning tasks, we define three error types to distinguish target selection failures from execution failures:

- **Target Execution Error:** The model selects the correct target but fails to execute the grasp or pick action properly.
- **No-Contact Grasp Error:** The model attempts to grasp but fails to contact any object.
- **Target Grounding Error:** The model selects the wrong object instead of the intended target.

For long-horizon procedural planning tasks, we define five error types. One of them is the Target Grounding Error (as above), and the remaining four are:

- **Manipulation Error:** Low-level execution of a subtask fails (e.g., pick, pull, or move).
- **Memory Error:** The model fails to retain relevant information from previous steps, affecting subtask completion.
- **Temporal Ordering Error:** Subtasks are executed in an incorrect order.
- **Redundant Repetition Error:** The model repeats an already completed action instead of progressing to the next subgoal.

A.2 Failure Analysis Results

Figure 7 and Figure 8 show the distribution of error types for the $\pi_{0.5}$ and X-VLA models, respectively. Overall, the two models exhibit broadly similar error patterns across the two capability dimensions, although the exact proportions vary.

Fine-Grained Spatial Reasoning. In these tasks, both models are primarily challenged by Target Grounding Errors, which account for over 70% of all errors. The remaining errors are mostly Target

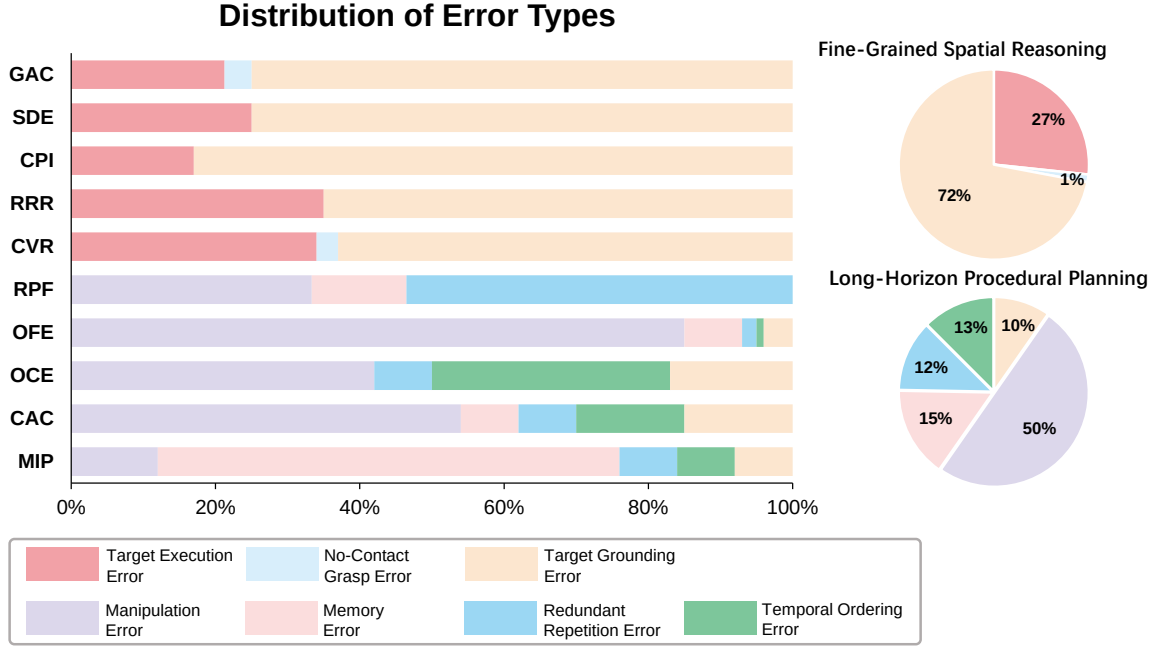


Figure 7: Distribution of error types for the $\pi_{0.5}$ model.

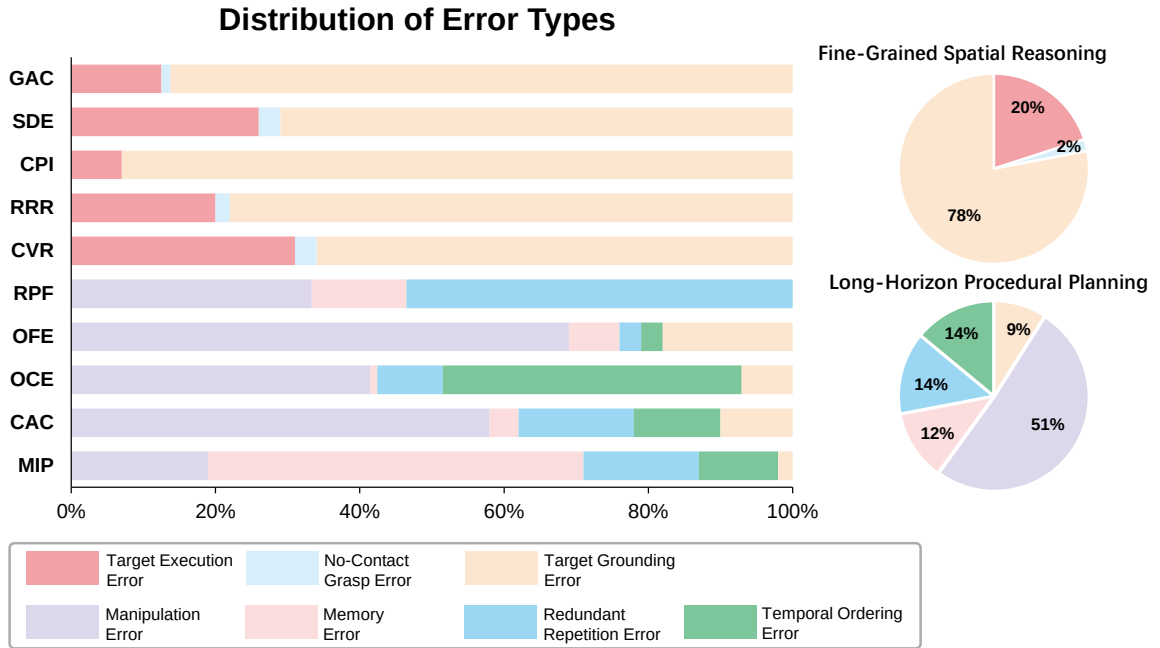


Figure 8: Distribution of error types for the X-VLA model.

Execution Errors, while No-Contact Grasp Errors occur very rarely. Among individual tasks, Canonical Position Indexing (CPI) exhibits the highest proportion of Target Grounding Errors, while Cross-View Reasoning (CVR) has a relatively larger share of Target Execution Errors compared with the other tasks. This indicates that the main challenge lies in insufficient spatial reasoning, while low-level execution errors are comparatively rare.

Long-Horizon Procedural Planning. Within this dimension, errors are diverse. Overall, Manipulation Errors account for the largest share, approximately 50%, while the remaining error types occur at roughly similar rates.

Repetitive Procedure Following (RPF) primarily exhibits Redundant Repetition Errors, suggesting that models lack effective tracking of task progress and may decide the next action based solely on the current visual observation. Order-Free Exe-

Table 5: Asset categories and the number of variants used in RoboSPA.

Asset	#Var.	Asset	#Var.	Asset	#Var.	Asset	#Var.
bottle	27	olive-oil	5	pencup	7	globe	2
bowl	7	drill	7	kitchenpot	7	trophy	5
cover	1	jam-jar	5	battery	6	notebook	3
plate	1	screwdriver	1	plasticbox	11	brush-pen	6
fluted-block	2	fork	1	tabletrashbin	11	rest	4
french-fries	4	knife	1	msg	6	glue	6
hamburg	6	apple	2	soy-sauce	5	cleaner	4
T_block	2	cabinet	1	vinegar	3	screen	4
shoe-box	1	box	1	steamer	3	speaker	6
tray	4	milk-box	4	boxdrink	4	fan	7
kettle	9	mug	13	vegetable	7	seal	6
pen	7	rack	1	paymentsign	4	milk-tea	6
dustbin	1	shoe	10	can	6	roller	3
plant-pot	5	wooden_box	1	electronicscale	5	fruit	7
dumbbell-rack	4	book	2	rubikscube	3	board	5
bookcase	4	microwave	2	displaystand	5	sauce-can	5
laptop	11	sand-clock	3	bread	7	skillet	4
oven	4	alarm-clock	6	breadbasket	5	soap	4
calculator	6	mouse	3	phone	5	block	7
microphone	4	stapler	7	phonestand	7	hydrating-oil	4
coaster	1	shampoo	7	remotecontrol	7	basket	4
hammer	1	bell	2	pillbottle	5	callbell	6
cup	13	candlestick	5	playingcards	3	tea-box	6
cup-with-liquid	1	dumbbell	7	smallshovel	4	coffee-box	7
tissue-box	7	teanet	5	brush	4	perfume	4
scanner	5	baguette	2	woodenmallet	5	keyboard	4
chips-tub	4	small-speaker	3	gong	6	whiteboard-eraser	1
pet-collar	4	switch	8	woodenblock	5	tooth-paste	1
table-tennis	2	toycar	6	waterer	8	mini-chalkboard	1
roll-paper	4	markpen	6	wineglass	5	plant	1

cution (OFE) and Composite Action Coordination (CAC) are dominated by Manipulation Errors, highlighting difficulties in low-level execution across multiple subgoals. Order-Constrained Execution (OCE) shows that, aside from Manipulation Errors, most remaining failures are Temporal Ordering Errors, indicating that models struggle to correctly sequence subtasks. Memory-Intensive Planning (MIP) is primarily affected by Memory Errors, indicating limitations in retaining task-relevant information across extended action sequences.

Summary. These task-specific error patterns demonstrate that different tasks in RoboSPA correspond to distinct failure modes, providing a benchmark with good task differentiation for diagnosing VLA model capabilities. They further suggest that future VLA models should strengthen spatial grounding, ensure precise low-level execution, implement effective progress tracking, and incorporate memory mechanisms to improve long-horizon tasks.

Table 6: Model size and training compute budget per base task. All experiments were conducted on NVIDIA RTX PRO 6000 GPUs.

Model	Params.	#GPUs	Time	GPU Hours.
RDT	1.2B	1	2.5 h	2.5
GO-1	2.8B	2	6 h	12
$\pi_{0.5}$	3.3B	2	6 h	12
X-VLA	0.9B	1	1.5 h	1.5

B Training and Evaluation Details

Training. For each base task, we use 50 trajectories per difficulty level, resulting in 250 trajectories across five levels. For each task variant, we generate 60 templates, with 50 used for training and 10 held out for evaluation. These templates are instantiated with task-specific object descriptions to form 100 distinct training instructions and 100 non-overlapping evaluation instructions. During training, one instruction is randomly sampled from the corresponding task-variant instruction pool for each trajectory.

To ensure a fair and controlled comparison, all

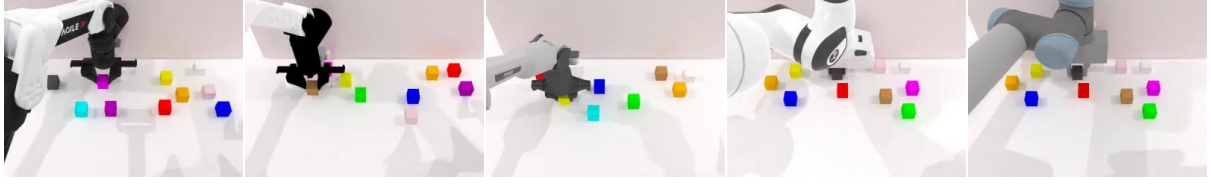


Figure 9: The five robotic embodiments supported by RoboSPA. From left to right: Aloha-AgileX, ARX-X5, Piper, Franka, and UR5.

baseline models are initialized from their officially released pretrained weights and fully fine-tuned on our dataset, with all parameters updated during training. We follow their official implementations, recommended environments, preprocessing pipelines, and evaluation settings, with only the training data and evaluation tasks replaced by RoboSPA. We adopt a unified training configuration across all models, using a batch size of 16 and training for 20,000 steps. All experiments are conducted on NVIDIA RTX PRO 6000 GPUs under consistent hardware settings. Table 6 summarizes the model sizes and training compute budgets per base task, including the number of parameters, GPU configuration, training time, and GPU hours for each evaluated baseline. Across all 56 base tasks and four baseline models, the total training budget is approximately 1,568 GPU-hours.

Evaluation. Each task variant is evaluated over 100 test trials, and the *Success Rate* is computed accordingly. To assess generalization ability, all evaluation instructions are strictly disjoint from those used during training, ensuring no overlap between training and test instructions. In addition to binary success, each task script records the completion status of individual actions or subtasks during execution. This enables step-level fine-grained evaluation for long-horizon tasks, where a rollout may fail the final goal while still completing part of the required procedure.

C Dataset Details

C.1 Object Asset Details

The object assets used in RoboSPA are collected from the RoboTwin-OD asset library (Chen et al., 2026a) and RMBench (Chen et al., 2026b). In total, RoboSPA includes 120 object categories and 581 object variants, covering diverse daily-use objects with different appearances, geometries, and semantic types. These assets are used to instantiate task scenes across both fine-grained spatial reason-

ing and long-horizon procedural planning tasks, as summarized in Table 5.

C.2 Multi-Embodiment Setting

RoboSPA supports multiple robotic embodiments, including Aloha-AgileX, ARX-X5, Piper, Franka, and UR5, as shown in Figure 9. These embodiments share the same task semantics, language instructions, object configurations, and success conditions, while differing in morphology, workspace, camera viewpoint, and execution behavior. This design enables evaluation of VLA models under diverse robotic configurations and tests whether models can generalize beyond a single embodiment.

C.3 Domain-Randomized Scenes

To evaluate robustness beyond clean environments, we collect additional trajectories under domain-randomized scenes, as shown in Figure 10. The task semantics and success conditions remain unchanged, while the visual and environmental conditions are varied.

Compared with clean scenes, the randomized setting introduces four types of variations: background appearance, tabletop clutter, table height, and lighting conditions. Background randomization exposes models to diverse visual contexts, while tabletop clutter introduces distractor objects. Table-height perturbation adds mild geometric variation, and lighting randomization changes illumination conditions.

D Basic Grounding Tasks

We additionally evaluate three basic grounding variants as lower-level references for spatial reasoning. These variants match the object categories and candidate counts of the corresponding RoboSPA tasks, namely SDE, CVR, and CPI, respectively. Color-Based Grounding relies only on color cues, Egocentric Position Grounding uses robot-view positions without cross-view transformation, and Simple Or-

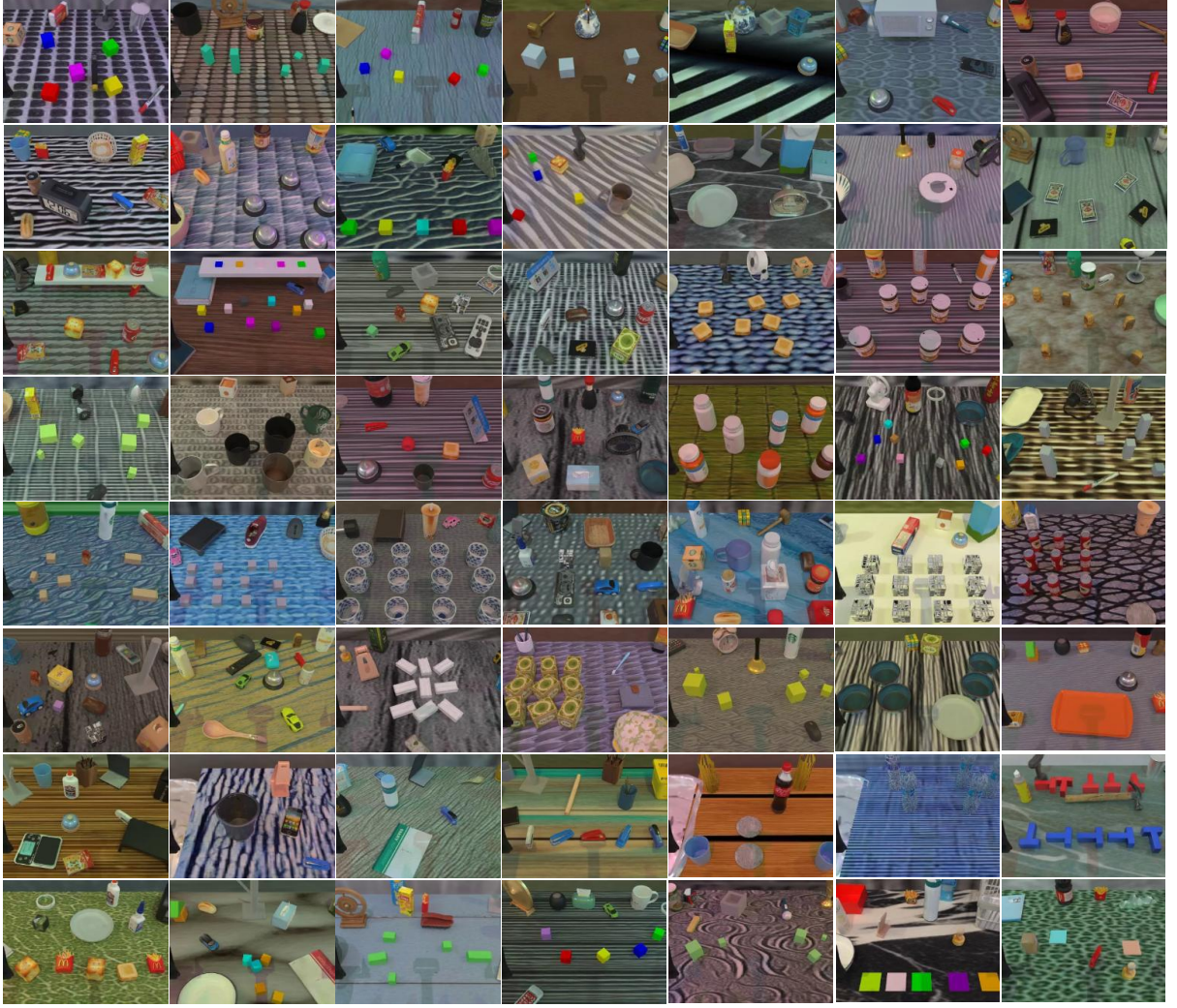


Figure 10: Examples of domain-randomized scenes in RoboSPA.

dinal Indexing fixes the counting directions to left-to-right and far-to-near, unlike CPI where indexing directions vary.

As shown in Table 7, these tasks achieve relatively high *Success Rates* across difficulty levels, indicating that basic grounding tasks may be insufficient to expose the reasoning limitations of current VLA models.

Table 7: Performance on basic spatial task variants.

Task	L1	L2	L3	L4	L5
Color-Based Grounding	80.6	77.8	79.8	71.2	74.0
Egocentric Position Grounding	67.8	69.4	74.8	72.2	72.4
Simple Ordinal Indexing	87.0	86.8	83.8	87.0	84.8

In contrast, RoboSPA introduces fine-grained spatial reasoning settings, such as distance estimation, cross-view reasoning, relational grounding, and canonical position indexing with more complex spatial layouts. These tasks provide a more chal-

lenging and diagnostic evaluation of spatial reasoning ability.

E Task List and Descriptions

Tables 8 and 9 provide the full list of base tasks in RoboSPA and their objectives. Each base task is instantiated into 5 difficulty variants, where fine-grained spatial reasoning tasks vary the number of candidate objects and long-horizon procedural planning tasks vary the number of procedural steps.

F Detailed Experimental Results

F.1 Statistical Uncertainty

We use a distinct seed for each evaluation episode, which determines the scene and object layout. Given the same checkpoint and seed, the simulated rollout is largely deterministic. To quantify the remaining finite-sample uncertainty, we report bootstrap 95% confidence intervals for the L5 results

in Table 10. The results show that the main performance trends remain unchanged after accounting for statistical uncertainty.

F.2 Multi-Embodiment Evaluation

To evaluate consistency across robotic platforms, we evaluate X-VLA on five embodiments. Due to computational resource constraints, we select one representative task from each of RoboSPA’s ten capability categories. For each task and embodiment, the model is independently trained and evaluated using data from that embodiment. Tasks are denoted by their category abbreviation and index (e.g., GAC-1), with the index corresponding to their order within the category in Tables 8 and 9.

As shown in Tables 11 and 12, absolute performance varies across embodiments, but the L5 success rate is lower than the L1 success rate for every evaluated task and embodiment. This demonstrates that RoboSPA exhibits a consistent difficulty trend across robotic embodiments.

F.3 Success Rates

To provide a complete view of model performance, we report the detailed per-task *Success Rates* for all evaluated models across the five difficulty levels. Tables 13–16 list the results of RDT, GO-1, $\pi_{0.5}$, and X-VLA on every task in RoboSPA. These tables complement the aggregated results in the main paper by showing how each model performs on individual task variants, making it easier to identify task-specific strengths and failure patterns.

Overall, the detailed results show that models generally perform better in settings with fewer distractor objects and shorter action sequences. However, performance drops become much sharper as difficulty increases, especially for tasks involving fine-grained spatial disambiguation, strict ordering, compositional execution, or memory-intensive planning.

F.4 Object-Normalized Target Accuracy

Tables 17–20 report the *Object-Normalized Target Accuracy* ($\mathcal{ONTA}$) of all evaluated models on fine-grained spatial reasoning tasks. These results help identify whether failures are caused by weak target disambiguation rather than low-level manipulation errors.

The $\mathcal{ONTA}$ results reveal clear differences in spatial grounding ability across models. RDT and GO-1 achieve poor scores, with many values below zero, indicating below-chance target selection

on numerous spatial tasks. $\pi_{0.5}$ and X-VLA perform better, but their scores still remain low on many variants and often stay close to the random-guessing baseline. These results suggest that current VLA models still have substantial room for improvement in fine-grained spatial reasoning, especially under increasing spatial ambiguity and scene complexity.

F.5 Progress Score

Tables 21–24 report the *Progress Scores* of all evaluated models on long-horizon procedural planning tasks. Unlike binary *Success Rate*, the *Progress Score* measures partial subtask completion during execution. These results provide a more fine-grained view of long-horizon failures by showing how much of the required procedure each model completes before failing.

The results highlight the limited long-horizon planning ability of current models. RDT and GO-1 obtain consistently low *Progress Scores*, with most tasks scoring below 50, suggesting that they often fail to complete even half of the required procedure. $\pi_{0.5}$ and X-VLA achieve higher *Progress Scores*, indicating better ability to complete intermediate subtasks. However, their scores still decline clearly as difficulty increases, showing that current VLA models remain far from robust long-horizon procedural planning.

G Task Visualizations

We provide visualizations for the most difficult variant (L5) of each task in RoboSPA, as shown in Figures 11–20. Each row shows representative rollout frames together with the task category, task name, and instruction, illustrating the scene layout, target objects, and required execution process.

Table 8: Task list and descriptions for fine-grained spatial reasoning. Category abbreviations: GAC = Geometric Attribute Cognition, SDE = Spatial Distance Estimation, CPI = Canonical Position Indexing, RRR = Referential Relational Reasoning, and CVR = Cross-View Reasoning.

Cat.	Task Name	Task Description
GAC	Pick Blocks Size	Pick the target block from multiple blocks based on the size.
GAC	Pick Blocks Height	Pick the target block from multiple blocks based on the height.
GAC	Pick Blocks Length	Pick the target block from multiple blocks based on the length.
GAC	Pick Blocks Area	Pick the target block from multiple blocks based on the base area.
SDE	Pick Blocks Distance	Pick the target block from multiple blocks using both color and relative distance cues.
SDE	Pick Mugs Distance	Pick the target mug from multiple mugs using both color and relative distance cues.
SDE	Pick Pill Bottles Distance	Pick the target pill bottle from multiple pill bottles using both color and relative distance cues.
SDE	Pick Mixed Objects Distance A	Pick the target object from mixed-object set A using both color and relative distance cues.
SDE	Pick Mixed Objects Distance B	Pick the target object from mixed-object set B using both color and relative distance cues.
CPI	Pick Blocks Canonical	Pick the target block according to its ordinal position among multiple blocks.
CPI	Pick Cups Canonical	Pick the target cup according to its ordinal position among multiple cups.
CPI	Pick Rubik Cubes Canonical	Pick the target Rubik cube according to its ordinal position among multiple Rubik cubes.
CPI	Pick Mixed Objects Canonical A	Pick the target object from mixed-object set A according to ordinal position.
CPI	Pick Mixed Objects Canonical B	Pick the target object from mixed-object set B according to ordinal position.
RRR	Pick Tea Box Relational	Pick the target tea box according to its relative position to another reference tea box.
RRR	Pick Cans Relational	Pick the target can according to its relative position to another reference can.
RRR	Pick Soaps Relational	Pick the target soap according to its relative position to another reference soap.
RRR	Pick Mixed Objects Relational A	Pick the target object in a mixed-object scene according to another reference object.
RRR	Pick Mixed Objects Relational B	Pick the target object in a mixed-object scene according to another reference object.
CVR	Pick Sauce Can Multi View	Pick up the target sauce can specified by its absolute position under a given viewpoint.
CVR	Pick Seals Multi View	Pick up the target seal specified by its absolute position under a given viewpoint.
CVR	Pick Breads Multi View	Pick up the target bread specified by its absolute position under a given viewpoint.
CVR	Pick Mixed Objects Multi View A	Pick up the target object from mixed-object group A specified by its absolute position under a given viewpoint.
CVR	Pick Mixed Objects Multi View B	Pick up the target object from mixed-object group B specified by its absolute position under a given viewpoint.

Table 9: Task list and descriptions for long-horizon procedural planning. Category abbreviations: RPF = Repetitive Procedure Following, OFE = Order-Free Execution, OCE = Order-Constrained Execution, CAC = Composite Action Coordination, and MIP = Memory-Intensive Planning.

Cat.	Task Name	Task Description
RPF	Press Stapler Repeat	Press the stapler repeatedly.
RPF	Lift Pot Repeat	Lift and put down the pot repeatedly.
RPF	Lift Fan Repeat	Lift the fan repeatedly.
RPF	Click Bell Repeat	Click the bell repeatedly.
OFE	Put Bottles Dustbin	Put multiple bottles into the dustbin without requiring a fixed execution order.
OFE	Move Blocks Apart	Move multiple blocks apart without requiring a fixed execution order.
OFE	Move Playing Cards Away	Move multiple playing cards away without requiring a fixed execution order.
OFE	Place Bowls Plates	Place multiple bowls on plates without requiring a fixed execution order.
OFE	Separate Fries Bread	Separate fries and bread into different areas without requiring a fixed execution order.
OFE	Rank Blocks Height	Arrange multiple blocks by height while allowing flexible intermediate execution order.
OFE	Rank Blocks Color	Arrange multiple blocks by color while allowing flexible intermediate execution order.
OFE	Rank Blocks Size	Arrange multiple blocks by size while allowing flexible intermediate execution order.
OCE	Hit Blocks Hammer Order	Hit multiple blocks with a hammer in the specified color order.
OCE	Click Objects Order	Click multiple target objects in the specified order.
OCE	Stamp Seals Order	Stamp multiple seals in the specified order.
OCE	Click Bell Clockwise Order	Click the bell and then execute the clockwise movement sequence in order.
OCE	Stack Blocks Color Order	Stack multiple blocks in the specified color order.
OCE	Stack Blocks Size Order	Stack multiple blocks in the specified size order.
OCE	Stack Blocks Length Order	Stack multiple blocks in the specified length order.
CAC	Place Phone Press Stapler	Execute a manipulation task involving phone placement, repeated pressing, and dumping.
CAC	Place Bottle Cup	Execute a manipulation task involving throwing bottles into a dustbin and placing cups on coasters.
CAC	Hang Mug Stack Blocks	Execute a manipulation task involving hanging a mug on a rack and stacking blocks from bottom to top.
CAC	Place Object Scale Click	Execute a manipulation task involving object placement, pressing, and returning objects to the table.
CAC	Place Burger Fries Click Bell	Execute a manipulation task involving placing a hamburger and fries on a tray, clicking a bell, and returning them.
CAC	Click Can Place Items	Execute a manipulation task involving interleaved can clicking with placing bread in a basket and a pill bottle on a pad.
CAC	Stamp Seals Press Stapler	Execute a manipulation task involving stamping left and right seals twice on their pads and pressing a stapler.
CAC	Click Bell Open Microwave Place Object	Execute a manipulation task involving interleaved bell clicking, opening a microwave, and placing an object left of another.
MIP	Observe Blocks Move Memory	First remember the colors of the display blocks, then move matching blocks on the table closer after the display blocks are hidden.
MIP	Observe Objects Click Memory	First remember the objects on the display platform from left to right, then click matching objects on the table in the same order after they are hidden.
MIP	Remember Color Cover	First remember the block colors, then uncover the blocks in the given color order after they are covered.
MIP	Remember Orientation Restore	First remember the orientations of the front T-shaped blocks, then adjust the rear T-shaped blocks to match them from left to right after they are hidden.
MIP	Press Stapler Memory	Remember how many times each stapler should be pressed and then press them.

Table 10: Bootstrap 95% confidence intervals for the average L5 *Success Rates* (%).

Model	Fine-Grained Spatial Reasoning Avg.	Long-Horizon Procedural Planning Avg.	Overall Avg.
RDT	6.1 [5.2, 7.1]	7.7 [6.8, 8.5]	6.9 [6.3, 7.5]
GO-1	10.6 [9.4, 11.8]	7.0 [6.2, 7.7]	8.8 [8.1, 9.5]
$\pi_{0.5}$	21.4 [19.8, 22.9]	23.1 [21.9, 24.3]	22.3 [21.3, 23.2]
X-VLA	23.9 [22.3, 25.5]	15.9 [14.8, 17.0]	19.9 [18.9, 20.8]

Table 11: Multi-embodiment evaluation of X-VLA on Fine-Grained Spatial Reasoning tasks.

Embodiment	GAC-1			SDE-5			CPI-4			RRR-3			CVR-4		
	L1	L3	L5	L1	L3	L5	L1	L3	L5	L1	L3	L5	L1	L3	L5
Aloha-AgileX	48	22	16	48	31	27	29	13	7	43	31	14	41	30	22
ARX-X5	51	18	11	36	29	26	32	14	7	50	24	13	28	11	13
Piper	48	34	16	18	20	17	40	18	10	40	18	20	50	44	39
Franka	22	12	8	14	10	10	38	18	8	48	18	14	12	8	7
UR5	74	43	25	52	41	34	67	35	26	58	45	38	59	55	49

Table 12: Multi-embodiment evaluation of X-VLA on Long-Horizon Procedural Planning tasks.

Embodiment	RPF-1			OFE-1			OCE-5			CAC-6			MIP-3		
	L1	L3	L5	L1	L3	L5	L1	L3	L5	L1	L3	L5	L1	L3	L5
Aloha-AgileX	49	30	27	79	55	28	99	67	18	54	37	24	100	8	0
ARX-X5	54	46	40	64	59	27	100	54	22	99	53	29	99	12	0
Piper	74	66	40	96	9	0	100	16	9	52	31	19	99	0	0
Franka	66	40	24	26	20	11	92	18	0	98	40	0	100	0	0
UR5	83	58	49	99	24	5	93	33	17	100	51	48	100	16	0

Table 13: RDT per-task *Success Rate* across difficulty levels.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	3	1	3	6	1	Put Bottles Dustbin	51	37	22	8	2
Pick Blocks Height	28	21	20	15	12	Move Blocks Apart	6	2	1	0	0
Pick Blocks Length	6	1	1	4	3	Move Playing Cards Away	8	5	0	0	0
Pick Blocks Area	3	1	1	5	3	Place Bowls Plates	48	15	12	6	2
Pick Blocks Distance	1	6	1	3	4	Separate Fries Bread	24	9	6	1	2
Pick Mugs Distance	15	17	12	13	10	Rank Blocks Height	6	1	0	1	0
Pick Pill Bottles Distance	25	15	14	13	13	Rank Blocks Color	14	6	1	0	0
Pick Mixed Objects Distance A	15	16	14	6	10	Rank Blocks Size	5	0	0	0	0
Pick Mixed Objects Distance B	20	13	14	13	10	Hit Blocks Hammer Order	40	4	0	0	0
Pick Blocks Canonical	3	5	4	1	5	Click Objects Order	20	0	0	0	0
Pick Cups Canonical	35	18	21	17	22	Stamp Seals Order	3	0	1	1	1
Pick Rubik Cubes Canonical	7	6	6	7	7	Click Bell Clockwise Order	5	0	0	0	0
Pick Mixed Objects Canonical A	11	4	5	1	5	Stack Blocks Color Order	11	0	0	0	0
Pick Mixed Objects Canonical B	19	6	5	6	4	Stack Blocks Size Order	9	0	0	0	0
Pick Tea Box Referential	7	2	5	6	3	Stack Blocks Length Order	7	0	0	0	0
Pick Cans Referential	2	1	7	5	6	Place Phone Press Stapler	18	6	1	0	0
Pick Soaps Referential	8	12	9	5	7	Place Bottle Cup	56	29	26	6	0
Pick Mixed Objects Referential A	2	2	7	7	4	Hang Mug Stack Blocks	10	0	0	0	0
Pick Mixed Objects Referential B	3	2	3	4	2	Place Object Scale Click	39	33	26	9	17
Pick Sauce Can Multi View	3	5	4	3	7	Place Burger Fries Click Bell	33	20	4	0	0
Pick Seals Multi View	3	5	5	4	4	Click Can Place Items	62	24	34	0	2
Pick Breads Multi View	3	5	2	3	0	Stamp Seals Press Stapler	2	3	0	0	0
Pick Mixed Objects Multi View A	1	5	2	3	4	Click Open Place	0	1	2	0	0
Pick Mixed Objects Multi View B	5	1	4	3	2	Observe Blocks Move Memory	0	0	0	0	0
Press Stapler Repeat	33	37	37	34	32	Observe Objects Click Memory	14	0	0	0	0
Lift Pot Repeat	76	68	63	56	68	Remember Color Cover	47	14	6	0	0
Lift Fan Repeat	39	33	26	9	17	Remember Orientation Restore	0	0	0	0	0
Click Bell Repeat	15	17	22	22	23	Press Stapler Memory	30	3	0	0	0

Table 14: GO-1 per-task *Success Rate* across difficulty levels.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	33	13	12	10	9	Put Bottles Dustbin	59	77	42	33	11
Pick Blocks Height	71	62	50	47	32	Move Blocks Apart	21	10	3	1	0
Pick Blocks Length	3	11	15	7	5	Move Playing Cards Away	52	39	33	13	17
Pick Blocks Area	4	3	2	0	0	Place Bowls Plates	37	23	21	13	5
Pick Blocks Distance	3	2	6	4	3	Separate Fries Bread	44	32	19	21	7
Pick Mugs Distance	15	14	14	8	9	Rank Blocks Height	34	3	0	0	0
Pick Pill Bottles Distance	36	8	15	15	26	Rank Blocks Color	41	14	8	5	2
Pick Mixed Objects Distance A	26	19	16	19	12	Rank Blocks Size	41	22	13	9	3
Pick Mixed Objects Distance B	21	17	14	8	8	Hit Blocks Hammer Order	36	11	3	3	1
Pick Blocks Canonical	16	11	8	6	7	Click Objects Order	62	11	6	4	3
Pick Cups Canonical	24	17	13	8	7	Stamp Seals Order	12	15	10	9	6
Pick Rubik Cubes Canonical	26	17	14	13	15	Click Bell Clockwise Order	61	48	34	21	12
Pick Mixed Objects Canonical A	21	12	6	6	4	Stack Blocks Color Order	54	12	3	1	0
Pick Mixed Objects Canonical B	23	15	7	4	3	Stack Blocks Size Order	38	11	1	0	0
Pick Tea Box Referential	21	17	18	9	7	Stack Blocks Length Order	30	7	5	0	0
Pick Cans Referential	16	12	4	14	8	Place Phone Press Stapler	48	14	14	5	1
Pick Soaps Referential	12	8	4	3	3	Place Bottle Cup	16	12	34	23	4
Pick Mixed Objects Referential A	20	18	13	15	9	Hang Mug Stack Blocks	8	2	0	0	0
Pick Mixed Objects Referential B	17	15	15	14	6	Place Object Scale Click	12	14	10	2	3
Pick Sauce Can Multi View	28	33	31	29	27	Place Burger Fries Click Bell	37	29	11	3	5
Pick Seals Multi View	18	15	14	12	11	Click Can Place Items	77	11	14	5	3
Pick Breads Multi View	21	12	9	27	15	Stamp Seals Press Stapler	24	17	2	0	0
Pick Mixed Objects Multi View A	16	8	4	18	18	Click Open Place	12	13	7	0	2
Pick Mixed Objects Multi View B	4	4	12	12	10	Observe Blocks Move Memory	3	0	0	0	0
Press Stapler Repeat	27	21	12	8	7	Observe Objects Click Memory	19	1	0	0	0
Lift Pot Repeat	61	53	83	84	77	Remember Color Cover	5	0	0	0	0
Lift Fan Repeat	23	17	17	21	11	Remember Orientation Restore	2	0	0	0	0
Click Bell Repeat	16	12	3	1	0	Press Stapler Memory	9	2	0	0	0

Table 15: $\pi_{0.5}$ per-task *Success Rate* across difficulty levels.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	58	27	20	17	13	Put Bottles Dustbin	95	85	81	74	61
Pick Blocks Height	67	54	48	30	24	Move Blocks Apart	73	51	24	35	18
Pick Blocks Length	22	5	11	4	3	Move Playing Cards Away	81	64	54	52	47
Pick Blocks Area	24	13	25	12	10	Place Bowls Plates	94	85	61	52	22
Pick Blocks Distance	16	16	16	10	11	Separate Fries Bread	92	84	58	59	36
Pick Mugs Distance	39	51	52	49	38	Rank Blocks Height	72	45	19	24	6
Pick Pill Bottles Distance	46	36	31	33	33	Rank Blocks Color	88	50	46	36	11
Pick Mixed Objects Distance A	33	37	27	29	23	Rank Blocks Size	91	66	37	15	8
Pick Mixed Objects Distance B	41	29	21	22	12	Hit Blocks Hammer Order	81	30	14	7	3
Pick Blocks Canonical	41	22	14	10	11	Click Objects Order	60	29	16	13	15
Pick Cups Canonical	61	37	40	41	45	Stamp Seals Order	40	40	30	36	35
Pick Rubik Cubes Canonical	51	35	33	36	44	Click Bell Clockwise Order	98	95	45	31	28
Pick Mixed Objects Canonical A	35	15	10	11	11	Stack Blocks Color Order	91	58	25	5	1
Pick Mixed Objects Canonical B	32	15	10	10	7	Stack Blocks Size Order	89	43	4	3	0
Pick Tea Box Referential	33	31	31	12	15	Stack Blocks Length Order	83	27	18	3	2
Pick Cans Referential	27	21	28	17	8	Place Phone Press Stapler	46	14	17	12	1
Pick Soaps Referential	53	44	38	19	10	Place Bottle Cup	97	79	69	67	30
Pick Mixed Objects Referential A	33	17	12	12	8	Hang Mug Stack Blocks	15	8	3	0	0
Pick Mixed Objects Referential B	20	21	18	18	8	Place Object Scale Click	19	36	56	6	5
Pick Sauce Can Multi View	47	48	43	42	42	Place Burger Fries Click Bell	87	89	57	14	34
Pick Seals Multi View	32	31	31	31	30	Click Can Place Items	78	46	47	14	18
Pick Breads Multi View	56	51	54	52	61	Stamp Seals Press Stapler	77	82	8	7	1
Pick Mixed Objects Multi View A	33	27	27	33	32	Click Open Place	50	22	37	5	11
Pick Mixed Objects Multi View B	38	29	30	22	23	Observe Blocks Move Memory	21	0	0	0	0
Press Stapler Repeat	79	41	42	36	45	Observe Objects Click Memory	97	35	3	0	0
Lift Pot Repeat	60	43	63	54	63	Remember Color Cover	100	48	11	2	0
Lift Fan Repeat	89	80	73	71	62	Remember Orientation Restore	20	0	1	0	0
Click Bell Repeat	93	84	87	95	90	Press Stapler Memory	32	3	0	0	0

Table 16: X-VLA per-task *Success Rate* across difficulty levels.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	48	27	22	23	16	Put Bottles Dustbin	79	64	55	31	28
Pick Blocks Height	88	81	65	56	55	Move Blocks Apart	86	74	81	17	31
Pick Blocks Length	43	30	21	28	14	Move Playing Cards Away	97	96	92	87	83
Pick Blocks Area	52	27	25	28	16	Place Bowls Plates	67	44	29	15	8
Pick Blocks Distance	39	44	33	29	33	Separate Fries Bread	78	87	71	79	66
Pick Mugs Distance	44	42	25	29	23	Rank Blocks Height	91	95	84	72	46
Pick Pill Bottles Distance	34	42	42	34	64	Rank Blocks Color	93	92	91	82	63
Pick Mixed Objects Distance A	42	40	36	34	26	Rank Blocks Size	97	88	65	32	24
Pick Mixed Objects Distance B	48	40	31	37	27	Hit Blocks Hammer Order	84	36	16	7	4
Pick Blocks Canonical	36	23	10	10	8	Click Objects Order	39	6	0	0	0
Pick Cups Canonical	38	23	20	19	11	Stamp Seals Order	34	71	65	65	31
Pick Rubik Cubes Canonical	26	18	14	5	12	Click Bell Clockwise Order	57	15	1	0	0
Pick Mixed Objects Canonical A	29	10	13	7	7	Stack Blocks Color Order	99	93	67	33	18
Pick Mixed Objects Canonical B	30	15	15	13	6	Stack Blocks Size Order	78	87	29	2	1
Pick Tea Box Referential	43	38	27	23	14	Stack Blocks Length Order	99	74	59	14	0
Pick Cans Referential	36	22	19	24	13	Place Phone Press Stapler	51	28	25	22	7
Pick Soaps Referential	43	30	31	15	14	Place Bottle Cup	98	44	19	16	6
Pick Mixed Objects Referential A	36	21	26	14	17	Hang Mug Stack Blocks	23	5	3	2	0
Pick Mixed Objects Referential B	44	34	23	16	13	Place Object Scale Click	57	22	19	5	2
Pick Sauce Can Multi View	54	53	62	57	59	Place Burger Fries Click Bell	98	59	18	12	4
Pick Seals Multi View	30	38	32	30	35	Click Can Place Items	54	41	37	34	24
Pick Breads Multi View	37	37	33	41	34	Stamp Seals Press Stapler	75	86	4	2	1
Pick Mixed Objects Multi View A	41	28	30	28	22	Click Open Place	3	2	0	0	0
Pick Mixed Objects Multi View B	29	27	33	36	33	Observe Blocks Move Memory	46	11	5	3	0
Press Stapler Repeat	49	35	30	29	27	Observe Objects Click Memory	18	0	0	0	0
Lift Pot Repeat	0	0	0	0	0	Remember Color Cover	100	48	8	0	0
Lift Fan Repeat	61	70	59	51	43	Remember Orientation Restore	18	6	0	0	0
Click Bell Repeat	29	24	20	19	20	Press Stapler Memory	41	1	0	0	0

Table 17: RDT per-task *Object-Normalized Target Accuracy (ONTA)* across difficulty levels. Fine-grained spatial reasoning tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	-94.0	-48.5	-29.3	-17.5	-18.8	Pick Mixed Objects Canonical A	-33.5	-15.2	-8.6	-11.4	-3.6
Pick Blocks Height	-44.0	-18.5	-6.7	-6.3	-5.6	Pick Mixed Objects Canonical B	-21.5	-12.8	-8.6	-5.8	-4.7
Pick Blocks Length	-88.0	-48.5	-32.0	-20.0	-16.4	Pick Tea Box Referential	-86.0	-47.0	-26.7	-12.8	-9.1
Pick Blocks Area	-94.0	-48.5	-32.0	-18.8	-16.4	Pick Cans Referential	-96.0	-48.5	-24.0	-14.0	-5.8
Pick Blocks Distance	-48.5	-12.8	-13.1	-9.1	-6.7	Pick Soaps Referential	-84.0	-32.0	-21.3	-14.0	-4.6
Pick Mugs Distance	-27.5	-10.7	-10.0	-4.4	-5.0	Pick Mixed Objects Referential A	-96.0	-47.0	-24.0	-11.6	-8.0
Pick Pill Bottles Distance	-12.5	-13.3	-7.5	-4.4	-1.5	Pick Mixed Objects Referential B	-94.0	-47.0	-29.3	-15.2	-10.3
Pick Mixed Objects Distance A	-27.5	-12.0	-7.5	-12.8	-5.0	Pick Sauce Can Multi View	-45.5	-26.7	-28.0	-29.3	-24.0
Pick Mixed Objects Distance B	-20.0	-16.0	-7.5	-4.4	-5.0	Pick Seals Multi View	-45.5	-26.7	-26.7	-28.0	-28.0
Pick Blocks Canonical	-45.5	-14.0	-9.7	-11.4	-3.6	Pick Breads Multi View	-45.5	-26.7	-30.7	-29.3	-33.3
Pick Cups Canonical	2.5	1.6	9.7	6.6	14.9	Pick Mixed Objects Multi View A	-48.5	-26.7	-30.7	-29.3	-28.0
Pick Rubik Cubes Canonical	-39.5	-12.8	-7.4	-4.6	-1.5	Pick Mixed Objects Multi View B	-42.5	-32.0	-28.0	-29.3	-30.7

Table 18: GO-1 per-task *Object-Normalized Target Accuracy* ($\mathcal{ONTA}$) across difficulty levels. Fine-grained spatial reasoning tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	-34.0	-30.5	-17.3	-12.5	-9.2	Pick Mixed Objects Canonical A	-18.5	-5.6	-7.4	-5.8	-4.7
Pick Blocks Height	42.0	43.0	33.3	33.8	18.4	Pick Mixed Objects Canonical B	-15.5	-2.0	-6.3	-8.0	-5.8
Pick Blocks Length	-94.0	-33.5	-13.3	-16.3	-14.0	Pick Tea Box Referential	-58.0	-24.5	-9.3	-9.2	-4.6
Pick Blocks Area	-92.0	-45.5	-30.7	-25.0	-20.0	Pick Cans Referential	-68.0	-32.0	-28.0	-3.2	-3.5
Pick Blocks Distance	-45.5	-17.6	-7.4	-8.0	-7.8	Pick Soaps Referential	-76.0	-38.0	-28.0	-16.4	-9.1
Pick Mugs Distance	-27.5	-14.7	-7.5	-10.4	-6.2	Pick Mixed Objects Referential A	-60.0	-23.0	-16.0	-2.0	-2.4
Pick Pill Bottles Distance	4.0	-22.7	-6.3	-2.0	13.7	Pick Mixed Objects Referential B	-66.0	-27.5	-13.3	-3.2	-5.8
Pick Mixed Objects Distance A	-11.0	-8.0	-5.0	2.8	-2.7	Pick Sauce Can Multi View	-8.0	10.7	8.0	5.3	2.7
Pick Mixed Objects Distance B	-18.5	-10.7	-7.5	-10.4	-7.3	Pick Seals Multi View	-23.0	-13.3	-14.7	-17.3	-18.7
Pick Blocks Canonical	-26.0	-6.8	-5.1	-5.8	-1.5	Pick Breads Multi View	-18.5	-17.3	-21.3	2.7	-13.3
Pick Cups Canonical	-14.0	0.4	0.6	-3.5	-1.5	Pick Mixed Objects Multi View A	-26.0	-22.7	-28.0	-9.3	-9.3
Pick Rubik Cubes Canonical	-11.0	0.4	1.7	2.1	7.3	Pick Mixed Objects Multi View B	-44.0	-28.0	-17.3	-17.3	-20.0

Table 19: $\pi_{0.5}$ per-task *Object-Normalized Target Accuracy* ($\mathcal{ONTA}$) across difficulty levels. Fine-grained spatial reasoning tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	16.0	-9.5	-6.7	-3.8	-4.4	Pick Mixed Objects Canonical A	2.5	-2.0	-2.9	-0.1	2.9
Pick Blocks Height	34.0	31.0	30.7	12.5	8.8	Pick Mixed Objects Canonical B	-2.0	-2.0	-2.9	-1.2	-1.5
Pick Blocks Length	-56.0	-42.5	-18.7	-20.0	-16.4	Pick Tea Box Referential	-34.0	-3.5	8.0	-5.6	4.4
Pick Blocks Area	-52.0	-30.5	0.0	-10.0	-8.0	Pick Cans Referential	-46.0	-18.5	4.0	0.4	-3.5
Pick Blocks Distance	-26.0	-0.8	4.0	-1.2	1.1	Pick Soaps Referential	6.0	16.0	17.3	2.8	-1.2
Pick Mugs Distance	8.5	34.7	40.0	38.8	27.7	Pick Mixed Objects Referential A	-34.0	-24.5	-17.3	-5.6	-3.5
Pick Pill Bottles Distance	19.0	14.7	13.8	19.6	21.8	Pick Mixed Objects Referential B	-60.0	-18.5	-9.3	1.6	-3.5
Pick Mixed Objects Distance A	-0.5	16.0	8.8	14.8	10.2	Pick Sauce Can Multi View	20.5	30.7	24.0	22.7	22.7
Pick Mixed Objects Distance B	11.5	5.3	1.3	22.0	-2.7	Pick Seals Multi View	-2.0	8.0	8.0	8.0	6.7
Pick Blocks Canonical	11.5	6.4	1.7	-1.2	2.9	Pick Breads Multi View	34.0	34.7	38.7	36.0	48.0
Pick Cups Canonical	41.5	24.4	31.4	33.6	40.0	Pick Mixed Objects Multi View A	-0.5	2.7	2.7	10.7	9.3
Pick Rubik Cubes Canonical	26.5	22.0	23.4	28.0	38.9	Pick Mixed Objects Multi View B	7.0	5.3	6.7	-4.0	-2.7

Table 20: X-VLA per-task *Object-Normalized Target Accuracy* ($\mathcal{ONTA}$) across difficulty levels. Fine-grained spatial reasoning tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Pick Blocks Size	-4.0	-9.5	-4.0	3.8	-0.8	Pick Mixed Objects Canonical A	-6.5	-8.0	0.6	-4.6	-1.5
Pick Blocks Height	76.0	71.5	53.3	45.0	46.0	Pick Mixed Objects Canonical B	-5.0	-2.0	2.9	2.1	-2.5
Pick Blocks Length	-14.0	-5.0	-5.3	10.0	-3.2	Pick Tea Box Referential	-14.0	7.0	2.7	7.6	3.3
Pick Blocks Area	4.0	-9.5	0.0	10.0	-0.8	Pick Cans Referential	-28.0	-17.0	-8.0	8.8	2.1
Pick Blocks Distance	8.5	32.8	23.4	20.1	25.6	Pick Soaps Referential	-14.0	-5.0	8.0	-2.0	3.3
Pick Mugs Distance	16.0	22.7	6.3	14.8	10.2	Pick Mixed Objects Referential A	-28.0	-18.5	1.3	-3.2	6.6
Pick Pill Bottles Distance	1.0	22.7	27.5	20.8	58.0	Pick Mixed Objects Referential B	-12.0	1.0	-2.7	-0.8	2.1
Pick Mixed Objects Distance A	13.0	20.0	20.0	20.8	13.7	Pick Sauce Can Multi View	31.0	37.3	49.3	42.7	45.3
Pick Mixed Objects Distance B	22.0	20.0	13.8	24.4	14.8	Pick Seals Multi View	-5.0	17.3	9.3	6.7	13.3
Pick Blocks Canonical	4.0	7.6	-2.9	-1.2	-0.4	Pick Breads Multi View	5.5	16.0	10.7	21.3	12.0
Pick Cups Canonical	7.0	7.6	8.6	8.9	2.9	Pick Mixed Objects Multi View A	11.5	4.0	6.7	4.0	-4.0
Pick Rubik Cubes Canonical	-11.0	1.6	1.7	-6.9	4.0	Pick Mixed Objects Multi View B	-6.5	2.7	10.7	14.7	10.7

Table 21: RDT per-task step *Progress Score* across difficulty levels. Step tables include long-horizon tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Press Stapler Repeat	33.0	37.0	39.3	36.5	35.2	Stack Blocks Color Order	11.0	50.0	33.7	25.0	20.2
Lift Pot Repeat	76.0	75.5	75.7	72.0	80.0	Stack Blocks Size Order	9.0	0.0	33.7	25.3	20.4
Lift Fan Repeat	39.0	41.5	41.3	29.8	34.0	Stack Blocks Length Order	7.0	50.0	34.0	25.3	20.4
Click Bell Repeat	15.0	21.5	28.7	26.5	28.6	Place Phone Press Stapler	18.0	40.0	44.0	49.3	38.2
Put Bottles Dustbin	51.0	56.0	48.3	33.8	33.6	Place Bottle Cup	56.0	52.5	52.7	44.3	33.6
Move Blocks Apart	6.0	7.0	12.7	11.0	8.0	Hang Mug Stack Blocks	10.0	9.0	17.7	13.0	9.6
Move Playing Cards Away	8.0	13.0	8.0	9.0	10.6	Place Object Scale Click	39.0	41.5	41.3	29.8	34.0
Place Bowls Plates	48.0	37.5	39.0	32.3	20.6	Place Burger Fries Click Bell	33.0	23.5	17.7	15.3	11.4
Separate Fries Bread	24.0	26.0	33.0	28.0	35.4	Click Can Place Items	62.0	55.5	75.7	55.0	65.0
Rank Blocks Height	6.0	50.5	40.0	37.8	31.4	Stamp Seals Press Stapler	2.0	5.0	1.3	0.8	3.0
Rank Blocks Color	14.0	53.0	44.3	37.5	31.4	Click Open Place	0.0	16.5	10.0	22.8	17.2
Rank Blocks Size	5.0	50.0	38.3	33.8	28.0	Observe Blocks Move Memory	0.0	0.0	0.0	0.0	0.0
Hit Blocks Hammer Order	40.0	14.5	9.3	6.5	7.2	Observe Objects Click Memory	14.0	7.0	5.3	6.3	5.8
Click Objects Order	20.0	23.5	22.7	17.5	0.0	Remember Color Cover	47.0	23.5	15.0	5.5	6.0
Stamp Seals Order	3.0	3.0	5.7	6.3	6.6	Remember Orientation Restore	0.0	0.5	0.7	0.0	0.2
Click Bell Clockwise Order	5.0	50.0	38.3	33.8	28.0	Press Stapler Memory	30.0	7.0	4.0	3.3	5.8

Table 22: GO-1 per-task step *Progress Score* across difficulty levels. Step tables include long-horizon tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Press Stapler Repeat	27.0	25.5	14.7	8.3	8.8	Stack Blocks Color Order	54.0	56.0	36.7	25.8	20.0
Lift Pot Repeat	61.0	54.0	92.0	92.8	91.6	Stack Blocks Size Order	38.0	16.5	5.0	4.3	4.2
Lift Fan Repeat	23.0	22.5	24.7	26.5	21.2	Stack Blocks Length Order	30.0	12.0	10.3	4.3	3.0
Click Bell Repeat	16.0	19.0	8.7	9.3	6.3	Place Phone Press Stapler	48.0	33.0	34.3	30.0	28.8
Put Bottles Dustbin	59.0	84.5	68.0	60.3	48.4	Place Bottle Cup	16.0	16.5	50.3	68.5	57.0
Move Blocks Apart	21.0	21.5	12.0	8.3	6.6	Hang Mug Stack Blocks	8.0	23.3	18.5	18.3	13.1
Move Playing Cards Away	52.0	43.5	45.0	20.5	21.0	Place Object Scale Click	12.0	19.0	14.3	6.8	8.2
Place Bowls Plates	37.0	36.0	35.0	26.8	20.8	Place Burger Fries Click Bell	37.0	32.5	13.7	5.0	6.6
Separate Fries Bread	44.0	37.5	26.7	27.8	14.6	Click Can Place Items	77.0	45.5	53.3	51.0	55.4
Rank Blocks Height	34.0	51.5	39.3	36.0	30.0	Stamp Seals Press Stapler	24.0	23.0	8.3	5.3	5.2
Rank Blocks Color	41.0	57.0	44.7	36.5	30.6	Click Open Place	12.0	13.0	7.0	0.0	5.8
Rank Blocks Size	41.0	61.0	47.3	40.5	34.0	Observe Blocks Move Memory	3.0	3.0	1.3	11.0	11.6
Hit Blocks Hammer Order	36.0	29.0	24.3	27.5	27.0	Observe Objects Click Memory	19.0	6.5	8.3	5.8	5.2
Click Objects Order	62.0	42.0	30.3	24.3	3.0	Remember Color Cover	5.0	0.0	0.0	0.3	0.0
Stamp Seals Order	12.0	29.5	24.0	23.3	21.4	Remember Orientation Restore	2.0	6.0	5.7	6.5	5.4
Click Bell Clockwise Order	61.0	52.0	38.3	30.8	25.2	Press Stapler Memory	9.0	6.5	9.3	9.5	7.6

Table 23: $\pi_{0.5}$ per-task step *Progress Score* across difficulty levels. Step tables include long-horizon tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Press Stapler Repeat	79.0	61.5	63.0	59.0	64.8	Stack Blocks Color Order	91.0	80.0	58.7	37.0	23.8
Lift Pot Repeat	60.0	64.5	77.0	71.8	74.8	Stack Blocks Size Order	89.0	75.0	49.3	36.3	25.0
Lift Fan Repeat	89.0	85.0	84.0	80.8	75.4	Stack Blocks Length Order	83.0	65.5	52.0	33.3	27.0
Click Bell Repeat	93.0	89.0	90.7	98.0	94.2	Place Phone Press Stapler	46.0	31.0	34.7	32.5	26.6
Put Bottles Dustbin	95.0	90.5	93.0	87.8	82.4	Place Bottle Cup	97.0	90.0	87.3	86.3	75.8
Move Blocks Apart	73.0	67.5	62.0	61.5	55.6	Hang Mug Stack Blocks	15.0	48.0	44.3	26.5	23.6
Move Playing Cards Away	81.0	78.5	77.3	81.5	77.8	Place Object Scale Click	19.0	50.5	56.0	22.8	30.2
Place Bowls Plates	94.0	91.5	78.0	69.5	50.4	Place Burger Fries Click Bell	87.0	90.0	78.0	61.8	72.0
Separate Fries Bread	92.0	88.5	83.0	86.8	74.4	Click Can Place Items	78.0	69.0	78.3	69.0	73.0
Rank Blocks Height	72.0	72.5	63.7	68.0	53.0	Stamp Seals Press Stapler	77.0	86.0	50.0	48.8	45.0
Rank Blocks Color	88.0	77.5	73.0	68.5	55.4	Click Open Place	50.0	48.5	55.0	30.8	37.4
Rank Blocks Size	91.0	84.0	69.7	57.5	47.8	Observe Blocks Move Memory	21.0	15.0	16.0	13.5	14.4
Hit Blocks Hammer Order	81.0	33.0	27.3	23.3	19.8	Observe Objects Click Memory	97.0	58.5	33.0	25.5	14.4
Click Objects Order	60.0	35.0	27.0	28.0	15.0	Remember Color Cover	100.0	50.5	20.3	12.0	7.0
Stamp Seals Order	40.0	50.5	43.0	43.6	46.6	Remember Orientation Restore	20.0	12.0	14.7	13.3	12.0
Click Bell Clockwise Order	98.0	96.5	52.7	40.8	38.6	Press Stapler Memory	32.0	10.0	9.7	8.8	9.4

Table 24: X-VLA per-task step *Progress Score* across difficulty levels. Step tables include long-horizon tasks only.

Task	L1	L2	L3	L4	L5	Task	L1	L2	L3	L4	L5
Press Stapler Repeat	49.0	42.5	40.3	38.8	36.6	Stack Blocks Color Order	99.0	96.5	79.0	54.5	37.8
Lift Pot Repeat	0.0	0.0	0.0	0.0	0.0	Stack Blocks Size Order	78.0	93.5	55.0	29.3	22.2
Lift Fan Repeat	61.0	79.5	72.0	71.8	66.2	Stack Blocks Length Order	99.0	87.5	73.7	41.8	24.2
Click Bell Repeat	29.0	28.5	27.0	23.3	24.8	Place Phone Press Stapler	51.0	45.5	46.7	43.5	43.8
Put Bottles Dustbin	79.0	80.0	78.3	71.0	74.0	Place Bottle Cup	98.0	71.0	61.3	49.0	40.0
Move Blocks Apart	86.0	80.5	89.7	65.0	71.2	Hang Mug Stack Blocks	23.0	51.0	51.7	41.3	31.4
Move Playing Cards Away	97.0	98.0	95.7	93.6	93.6	Place Object Scale Click	57.0	35.0	30.3	27.3	20.0
Place Bowls Plates	67.0	62.5	52.7	39.3	30.4	Place Burger Fries Click Bell	98.0	78.5	55.7	57.3	49.8
Separate Fries Bread	78.0	92.5	87.3	89.5	82.2	Click Can Place Items	54.0	64.5	67.7	69.3	64.2
Rank Blocks Height	91.0	97.5	93.3	91.0	83.0	Stamp Seals Press Stapler	75.0	89.0	61.0	49.3	42.8
Rank Blocks Color	93.0	96.5	95.0	92.5	90.0	Click Open Place	15.0	19.5	18.0	24.8	28.8
Rank Blocks Size	97.0	94.5	83.3	71.0	67.0	Observe Blocks Move Memory	46.0	29.5	27.7	27.3	19.4
Hit Blocks Hammer Order	84.0	57.5	46.7	40.5	30.8	Observe Objects Click Memory	18.0	4.5	5.0	2.8	2.6
Click Objects Order	39.0	20.0	10.3	6.5	0.0	Remember Color Cover	100.0	50.0	18.3	6.5	4.0
Stamp Seals Order	34.0	77.0	78.3	78.3	53.2	Remember Orientation Restore	18.0	19.5	7.0	6.3	5.6
Click Bell Clockwise Order	57.0	26.0	14.7	9.8	7.2	Press Stapler Memory	41.0	6.5	3.0	4.3	4.8

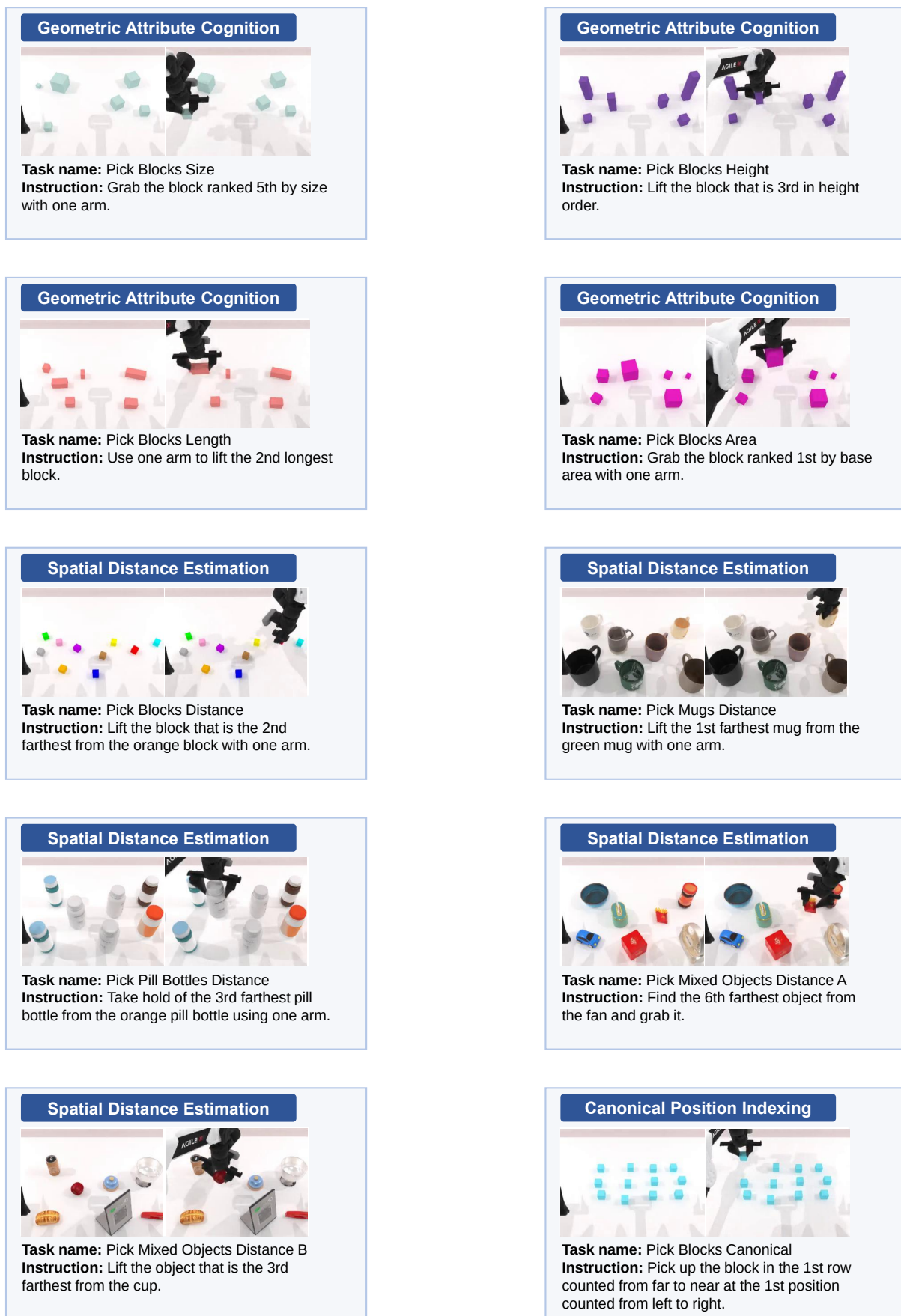
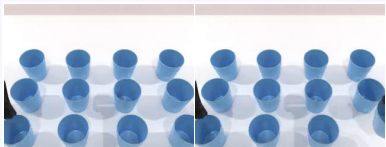


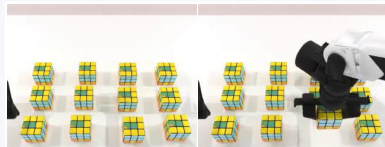
Figure 11: Fine-grained spatial reasoning task examples in RoboSPA (Part 1).

Canonical Position Indexing



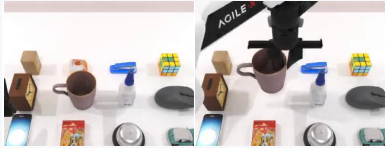
Task name: Pick Cups Canonical
Instruction: Find the 4th cup in the 1st row counted from near to far, with positions counted from left to right, and pick it up.

Canonical Position Indexing



Task name: Pick Rubik Cubes Canonical
Instruction: Pick up the 2nd rubik's cube counted from right to left in the 1st row counted from near to far.

Canonical Position Indexing



Task name: Pick Mixed Objects Canonical A
Instruction: Pick up the object in the 2nd row counted from far to near at the 2nd position counted from left to right.

Canonical Position Indexing



Task name: Pick Mixed Objects Canonical B
Instruction: Use one arm to take the target object in the 2nd row counted from far to near at the 1st position counted from right to left.

Referential Relational Reasoning



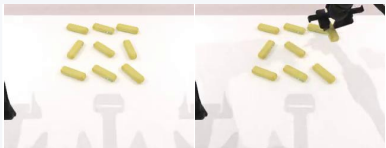
Task name: Pick Tea Box Relational
Instruction: Front is the side farther from the robot. Grab the nearest one to the front-right of the center tea box using one arm.

Referential Relational Reasoning



Task name: Pick Cans Relational
Instruction: The side farther from the robot is front. Use an arm to lift the nearest one directly at the back of the middle-left can.

Referential Relational Reasoning



Task name: Pick Soaps Relational
Instruction: The side farther from the robot is front. Pick the nearest one directly in front of the back-right soap bar.

Referential Relational Reasoning



Task name: Pick Mixed Objects Relational A
Instruction: Back means the side nearer to the robot. Take the closest object to the front-left of the back-middle object.

Referential Relational Reasoning



Task name: Pick Mixed Objects Relational B
Instruction: Back means the side nearer to the robot. Lift the nearest object directly in front of the middle-left object.

Cross-View Reasoning



Task name: Pick Sauce Can Multi View
Instruction: From the robot's side of the table, pick up the leftmost object.

Figure 12: Fine-grained spatial reasoning task examples in RoboSPA (Part 2).

Cross-View Reasoning



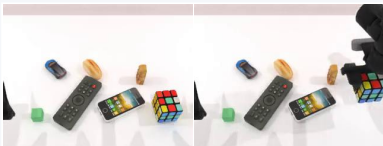
Task name: Pick Seals Multi View
Instruction: Seen from the opposite side of the table, take the rightmost object with an arm.

Cross-View Reasoning



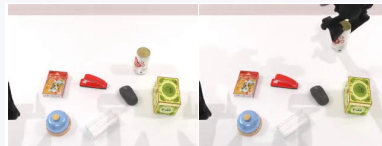
Task name: Pick Breads Multi View
Instruction: Viewed from the right side, use one arm to pick up the leftmost object.

Cross-View Reasoning



Task name: Pick Mixed Objects Multi View A
Instruction: From the left side of the table, pick up the frontmost object.

Cross-View Reasoning



Task name: Pick Mixed Objects Multi View B
Instruction: With the camera on the opposite side of the table, grab the backmost object with an arm.

Figure 13: Fine-grained spatial reasoning task examples in RoboSPA (Part 3).

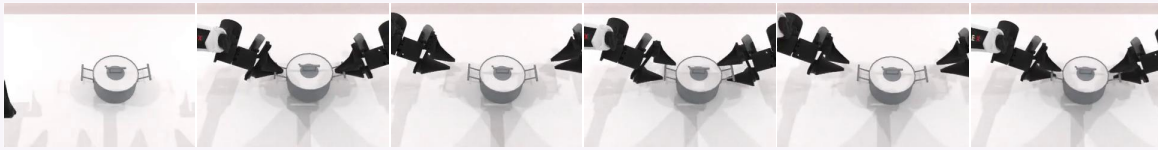
Repetitive Procedure Following



Task name: Press Stapler Repeat

Instruction: Press the blue stapler with shiny silver parts directly 5 times.

Repetitive Procedure Following



Task name: Lift Pot Repeat

Instruction: Lift and put down the cooking kitchen pot with side grips with arms 5 times.

Repetitive Procedure Following



Task name: Lift Fan Repeat

Instruction: Lift the white fan and set it down 5 times.

Repetitive Procedure Following



Task name: Click Bell Repeat

Instruction: Press the top center of the compact tabletop bell 5 times.

Order-Free Execution

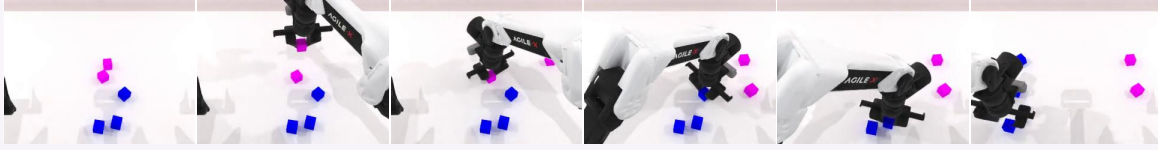


Task name: Put Bottles Dustbin

Instruction: Lift the five beverage bottles and drop them into the plastic dustbin.

Figure 14: Long-horizon procedural planning task examples in RoboSPA (Part 1).

Order-Free Execution



Task name: Move Blocks Apart

Instruction: Set every magenta block on the right and every blue block on the left.

Order-Free Execution



Task name: Move Playing Cards Away

Instruction: Move all boxes of playing cards to the left and right.

Order-Free Execution



Task name: Place Bowls Plates

Instruction: Stack each medium-sized bowl for holding food one by one onto the dish plate.

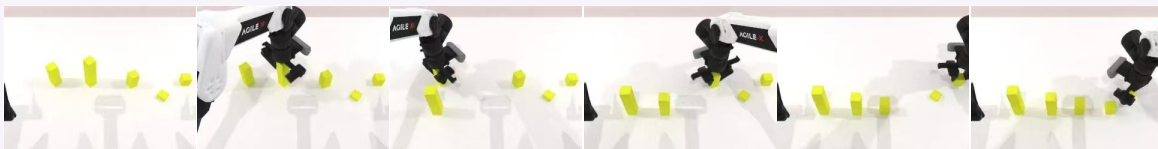
Order-Free Execution



Task name: Separate Fries Bread

Instruction: Place every bread at the front and every fries at the back.

Order-Free Execution

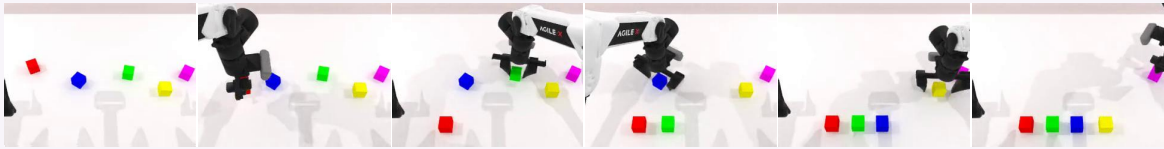


Task name: Rank Blocks Height

Instruction: Order the blocks in height order, tallest left to shortest right.

Figure 15: Long-horizon procedural planning task examples in RoboSPA (Part 2).

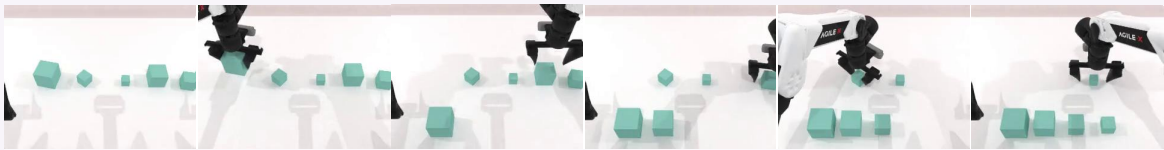
Order-Free Execution



Task name: Rank Blocks Color

Instruction: Order red, green, blue, yellow, and purple blocks from left to right.

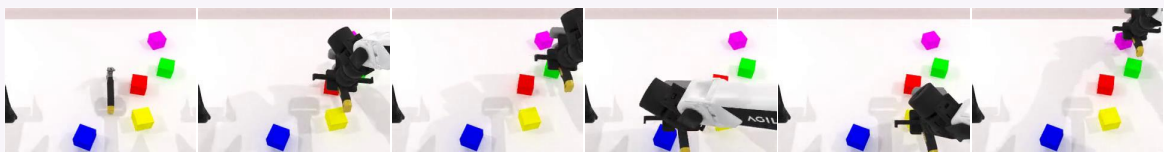
Order-Free Execution



Task name: Rank Blocks Size

Instruction: Sort the blocks by size from left to right, largest to smallest.

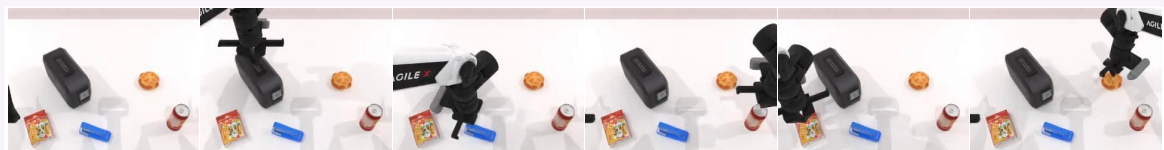
Order-Constrained Execution



Task name: Hit Blocks Hammer Order

Instruction: Lift the grippy handle hammer using the right arm to hit the red, green, blue, yellow, then purple block.

Order-Constrained Execution



Task name: Click Objects Order

Instruction: Press the rectangular alarm-clock, followed by the blue stapler, the red beverage can with curved top, the playingcards storage box, and the soft golden bread.

Order-Constrained Execution

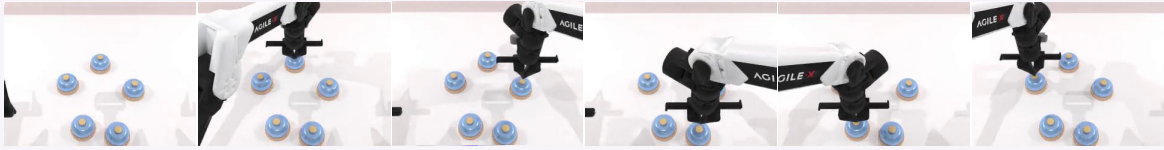


Task name: Stamp Seals Order

Instruction: Grab the brown seal, stamp on Olive, Coral, Teal, Tan, then Beige.

Figure 16: Long-horizon procedural planning task examples in RoboSPA (Part 3).

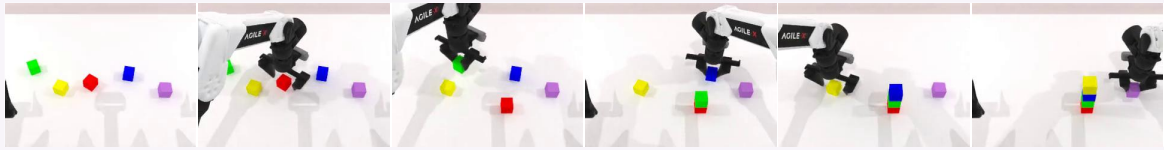
Order-Constrained Execution



Task name: Click Bell Clockwise Order

Instruction: Click all five bells starting from the frontmost one and moving clockwise.

Order-Constrained Execution



Task name: Stack Blocks Color Order

Instruction: Stack the blocks at the center from bottom to top as red, green, blue, yellow, and purple.

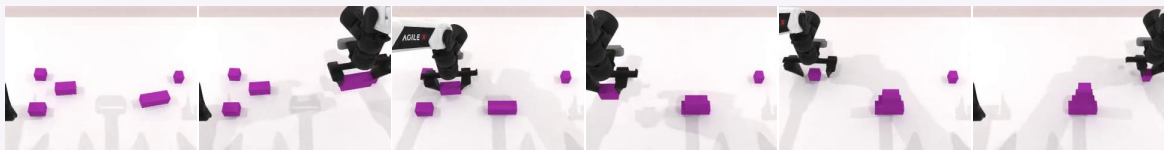
Order-Constrained Execution



Task name: Stack Blocks Size Order

Instruction: Stack the blocks at the center from largest to smallest, bottom to top.

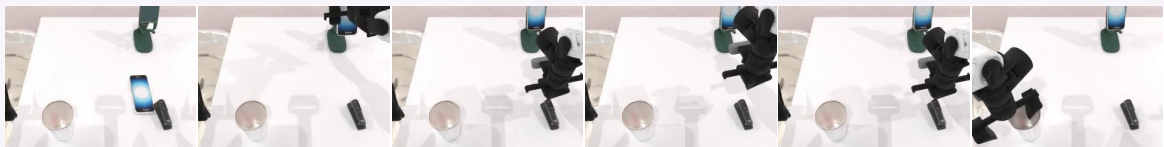
Order-Constrained Execution



Task name: Stack Blocks Length Order

Instruction: Use the arms to build a center stack from longest to shortest, bottom to top.

Composite Action Coordination

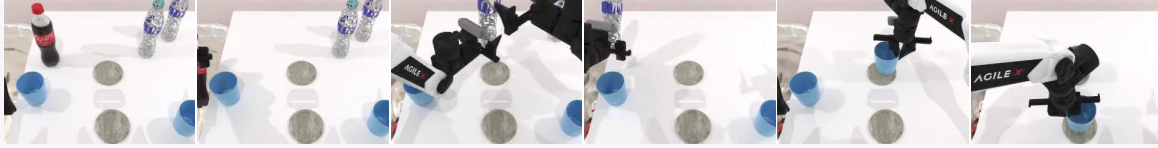


Task name: Place Phone Press Stapler

Instruction: Place the sleek phone on the flat base phone stand, press the curved black stapler thrice, then empty tabletop trash bin into the trash bin.

Figure 17: Long-horizon procedural planning task examples in RoboSPA (Part 4).

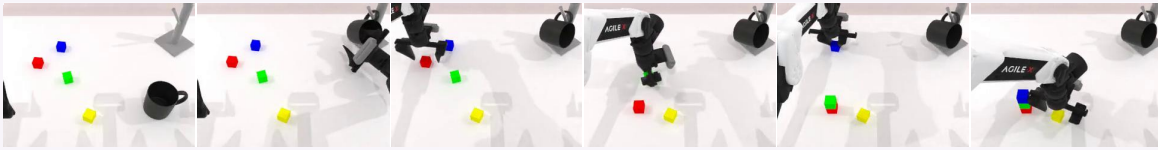
Composite Action Coordination



Task name: Place Bottle Cup

Instruction: Move three bottles in the trash bin one after another, and place the two plastic cups for drinks on the two flat round wooden coasters one by one.

Composite Action Coordination



Task name: Hang Mug Stack Blocks

Instruction: First hang the solid black ceramic mug on the metallic rack with smooth finish, then stack red block, green block, blue block, and yellow block in order from bottom to top.

Composite Action Coordination



Task name: Place Object Scale Click

Instruction: Start by placing the bell with black flat base on the black and white electronic scale, then put the toy car made of plastic material on the smooth textured black display stand, and press the box with cards inside, then put the bell with black flat base back on the table, and put the toy car made of plastic material back on the table.

Composite Action Coordination



Task name: Place Burger Fries Click Bell

Instruction: Deposit the orange hamburger with white underside on the orange tray, then deposit palm-sized red fries container on the orange tray, tap the plastic and metal bell once, then place both back.

Composite Action Coordination

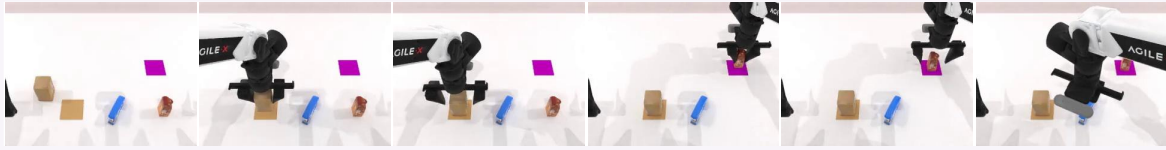


Task name: Click Can Place Items

Instruction: Press the brown can, move the bread block with light patches to the beige plastic breadbasket, click the brown can, set the bottle with large orange label on the pad, and finally press the brown can.

Figure 18: Long-horizon procedural planning task examples in RoboSPA (Part 5).

Composite Action Coordination



Task name: Stamp Seals Press Stapler

Instruction: Tap the left seal twice on left, tap the right seal twice on right, then press the stapler with curved blue top once.

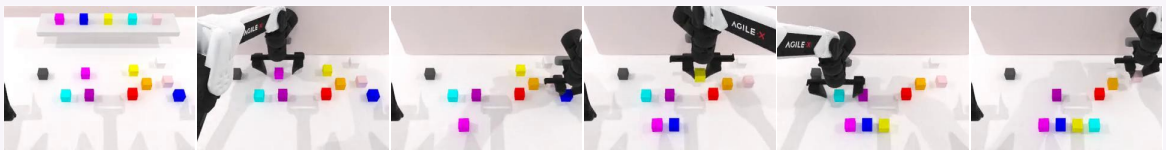
Composite Action Coordination



Task name: Click Bell Open Microwave Place Object

Instruction: Ring the plastic and metal bell, open the countertop microwave, ring the plastic and metal bell again, position the coffee-box with smooth card board left of the dark gray mouse, and ring the plastic and metal bell.

Memory-Intensive Planning



Task name: Observe Blocks Move Memory

Instruction: Observe the colors of the blocks on the display platform, then after they are hidden, move matching blocks on the table closer to you.

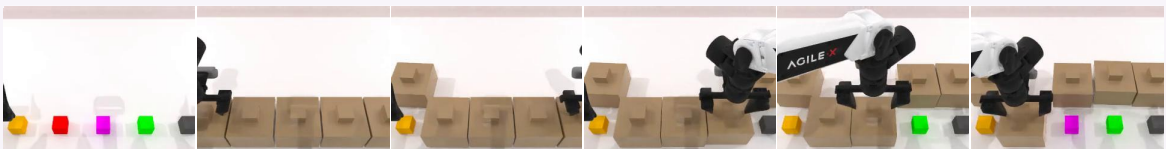
Memory-Intensive Planning



Task name: Observe Objects Click Memory

Instruction: Observe the objects on the display platform from left to right, then after they are hidden, click the matching objects on the table in the same order.

Memory-Intensive Planning



Task name: Remember Color Cover

Instruction: Memorize the colors of the blocks, then once they are covered, uncover the blocks in the order of orange, black, green, magenta, red.

Figure 19: Long-horizon procedural planning task examples in RoboSPA (Part 6).

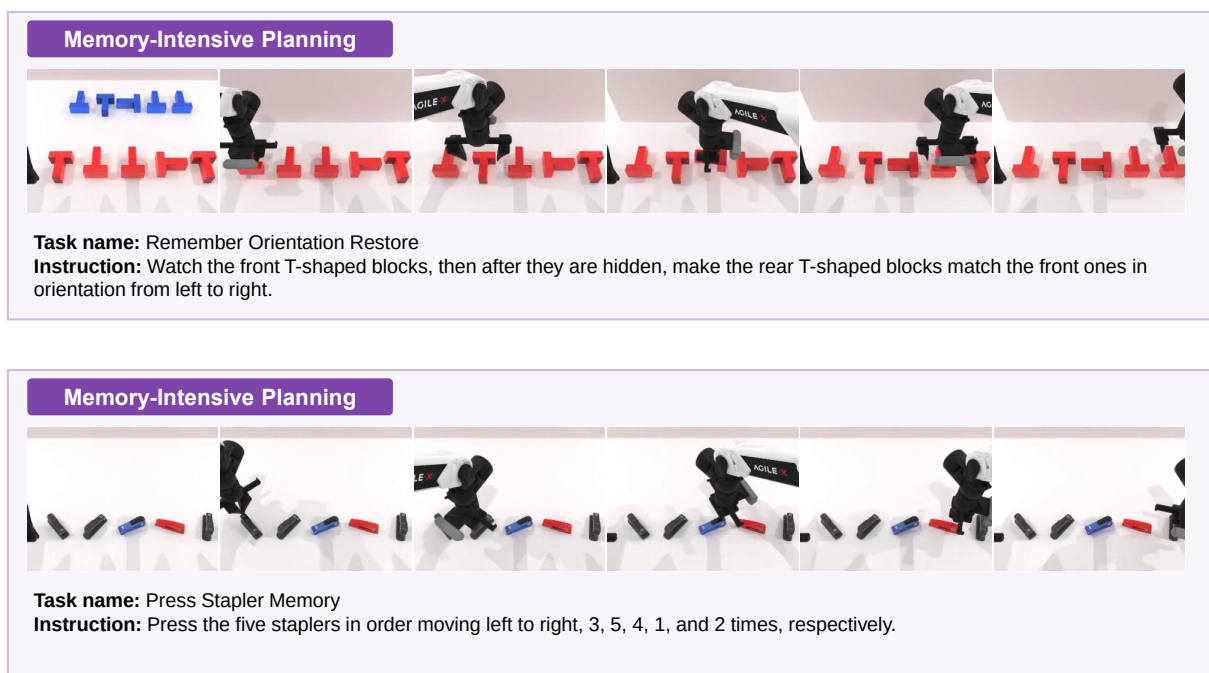


Figure 20: Long-horizon procedural planning task examples in RoboSPA (Part 7).